# Standard Model

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**Abstract**

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# 1 Gauge Redundancy

In order to describe a relativistic QFT containing massless vector particles, we use a gauge theory. For a vector field $A_\mu$, the familiar redundancy is

$$A_\mu \to A_\mu + \partial_\mu \alpha, \tag{1}$$

where $\alpha(x)$ is an arbitrary function. We now derive the origin of this redundancy from one-particle representations of the Poincare group.

## 1.1 One-particle representations

A relativistic one-particle state furnishes a unitary representation of the Poincare group. A Poincare transformation acts as

$$x'^\mu = \Lambda^\mu_\nu x^\nu + a^\mu, \quad \Lambda \in \mathrm{SO}(3,1). \tag{2}$$

Writing $g = g(\Lambda, a)$, the group law is

$$g(\Lambda_1, a_1) g(\Lambda_2, a_2) = g(\Lambda_1 \Lambda_2, \Lambda_1 a_2 + a_1), \tag{3}$$

and hence

$$\begin{aligned} g^{-1}(\Lambda, a) &= g(\Lambda^{-1}, -\Lambda^{-1} a), \\ g(\Lambda, a) &= g(1, a) g(\Lambda, 0). \end{aligned} \tag{4}$$

Thus we are able to separate the translation and Lorentz transformation parts and each part can be written in terms of generators and exp maps. Translations and Lorentz transformations are generated by

$$g(1,a) = \exp\big(iP^\mu a_\mu\big), \quad g(\Lambda,0) = \exp\Big(-\frac{i}{2}\omega_{\mu\nu}J^{\mu\nu}\Big). \tag{5}$$

The algebra of the generators is:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(\eta^{\mu\sigma}J^{\nu\rho} + \eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - \eta^{\nu\sigma}J^{\mu\rho}), \tag{6}$$

$$[P^\mu, J^{\rho\sigma}] = i(\eta^{\mu\rho}P^\sigma - \eta^{\mu\sigma}P^\rho), \quad [P^\mu, P^\nu] = 0. \tag{7}$$

Since the momentum components commute, we label one-particle states by their momentum and by an internal label $\sigma$:

$$P^\mu \ |p,\sigma\rangle = p^\mu \ |p,\sigma\rangle. \tag{8}$$

Under translation they transforms as a phase factor and under a Lorentz transformation,

$$U(\Lambda) \ |p,\sigma\rangle = \sum_{\sigma'} C_{\sigma\sigma'}(\Lambda,p) \ |\Lambda p,\sigma'\rangle. \tag{9}$$

Lorentz transformations preserves the following quantities of a state (this means that the LT matrix commute with these operators):

$$p^2 = m^2, \quad W^2, \quad \text{sign}(p^0), \tag{10}$$

where $W^\mu$ is the Pauli-Lubanski vector. This gives us an approach to classify the representations of the Lorentz group cause states with different values of the conserved quantities cannot be connected by a Lorentz transformation.

First of all, the orbits of the Lorentz group in momentum space are classified by the mass $m$ and the sign of $p^0$, physically we don't consider negative energy and 0 energy correspond to vaccum, the left over orbits are classified with different mass. For each orbit, choose a representative momentum $k^\mu$. For every $p$ on the same orbit, choose a standard Lorentz transformation $L(p)$ satisfying

$$p^\mu = L^\mu_\nu(p)k^\nu. \tag{11}$$

Then define

$$|p,\sigma\rangle \coloneqq N_p U(L(p)) \ |k,\sigma\rangle. \tag{12}$$

where we leave $\sigma$ as the index of some unknown parameters and $N_p$ is a normalization factor. Under a general Lorentz transformation,

$$\begin{aligned} U(\Lambda)|p,\sigma\rangle &= N_p U(\Lambda L(p))|k,\sigma\rangle \\ &= N_p U(L(\Lambda p))U(L^{-1}(\Lambda p)\Lambda L(p))|k,\sigma\rangle. \end{aligned} \tag{13}$$

The transformation

$$W(\Lambda,p) \coloneqq L^{-1}(\Lambda p)\Lambda L(p) \tag{14}$$

leaves $k$ invariant:

$$W^\mu_\nu k^\nu = k^\mu. \tag{15}$$

It therefore belongs to the **Wigner's little group** of $k$. Its action can be defined as some representation of the little group on the internal label $\sigma$:

$$U(W)|k,\sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}(W)|k,\sigma'\rangle, \tag{16}$$

Consequently,

$$U(\Lambda)\ |p,\sigma\rangle = \frac{N_p}{N_{\Lambda p}} \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda,p))\ |\Lambda p,\sigma'\rangle. \tag{17}$$

Thus, classifying one-particle states reduces to classifying representations of the little group.

### 1.1.1 Massive particles

For $m > 0$, choose

$$k^\mu = (m,0,0,0). \tag{18}$$

Spatial rotations leave $k^\mu$ invariant, so the little group is SO(3) (or its double cover SU(2)). Its finite-dimensional irreducible representations are labeled by

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, .... \tag{19}$$

> **Remark**
>
> In fact SO(3) doesn't have spin half representation, but SU(2) does. The little group is actually SU(2) which is the double cover of SO(3). This may relate to the fact that our symmetry group is not really SO(3,1).

### 1.1.2 Massless particles

For $m = 0$, choose

$$k^\mu = E(1,0,0,1). \tag{20}$$

The little group is generated by rotations about the $z$-axis $J^3$ and two generators

$$T_1 := K^1 - J^2, \quad T_2 := K^2 + J^1, \tag{21}$$

with

$$J^i := \frac{1}{2}\varepsilon^{ijk}J^{jk}, \quad K^i := J^{0i}. \quad \varepsilon^{123} = 1. \tag{22}$$

In the four-vector representation, we can easily show that the generators generate the Wigner's little group of $k$:

$$(J^{\mu\nu})^\rho_\sigma = i(\delta^\nu_\sigma \eta^{\mu\rho} - \delta^\mu_\sigma \eta^{\nu\rho}). \tag{23}$$

For the chosen $k$, one finds

$$\begin{aligned} K^1 k = J^2 k = iE(0,1,0,0), \\ K^2 k = -J^1 k = iE(0,0,1,0), \end{aligned} \tag{24}$$

and therefore $T_1 k = T_2 k = 0$.

Their algebra is

$$[J^3, T_1] = iT_2, \quad [J^3, T_2] = -iT_1, \quad [T_1, T_2] = 0. \tag{25}$$

This is the Lie algebra of ISO(2): $J^3$ generates rotations in the plane, while $T_1$ and $T_2$ generate translations. Because $T_1$ and $T_2$ commute, choose simultaneous eigenstates to label the states in the representation:

$$T_1\ |a,b\rangle = a\ |a,b\rangle, \quad T_2\ |a,b\rangle = b\ |a,b\rangle. \tag{26}$$

Using the Baker-Campbell-Hausdorff formula,

$$\begin{aligned} e^{i\theta J^3} T_1 e^{-i\theta J^3} &= T_1 \cos\theta - T_2 \sin\theta, \\ e^{i\theta J^3} T_2 e^{-i\theta J^3} &= T_2 \cos\theta + T_1 \sin\theta. \end{aligned} \tag{27}$$

Therefore

$$e^{-i\theta J^3} \ |a,b\rangle = |a\cos\theta - b\sin\theta, a\sin\theta + b\cos\theta\rangle. \tag{28}$$

Thus we can build a representation of the little group by taking a circle of states with the same radius in the $(a,b)$ plane. The radius

$$\rho := \sqrt{a^2+b^2} \tag{29}$$

is invariant. If $\rho \neq 0$, the representation contains a continuous circle of states and is infinite-dimensional. An equivalent way of showing that we have infinite number of state is by defining

$$T_+ := T_1 + iT_2, \quad T_- := T_1 - iT_2, \tag{30}$$

this gives

$$[J^3, T_+] = T_+, \quad [J^3, T_-] = -T_-, \quad [T_+, T_-] = 0. \tag{31}$$

If

$$J^3|h\rangle = h|h\rangle, \tag{32}$$

then

$$\begin{aligned} J^3 T_+ |h\rangle &= (h+1)T_+|h\rangle, \\ J^3 T_- |h\rangle &= (h-1)T_-|h\rangle. \end{aligned} \tag{33}$$

Unlike ordinary angular momentum, this ladder does not terminate when $\rho \neq 0$. Such representations are the continuous-spin representations which is not observed in the physical world. For the familiar finite-helicity particles, we impose the trivial action

$$J^3|p,h\rangle = h|p,h\rangle, \quad T_i|p,h\rangle = 0. \tag{34}$$

The pure translation part of the little group may be represented by

$$W^\mu{}_\nu(\alpha,\beta) = \begin{pmatrix} 1+\zeta & \alpha & \beta & -\zeta \\ \alpha & 1 & 0 & -\alpha \\ \beta & 0 & 1 & -\beta \\ \zeta & \alpha & \beta & 1-\zeta \end{pmatrix}, \quad \zeta := \frac{\alpha^2+\beta^2}{2}. \tag{35}$$

Indeed,

$$W^\mu{}_\nu k^\nu = k^\mu. \tag{36}$$

Though we wish the translation part of the little group to act trivially on the quantum state, yet if we define a spin 1 theory that have local degrees of freedom that transforms as a 4 vector, then it must have polarization internal degrees of freedom that transforms non-trivially under the translation part of the little group. To resolve this, we need to identify different polarization vectors and understand them as the same physical state. This is the origin of gauge redundancy.

Take the two transverse polarization vectors (we care about transverse polarization because the EoMs of massless vector field implies that the polarization vector must be transverse to the momentum vector)

$$\varepsilon_1^\mu = (0,1,0,0), \quad \varepsilon_2^\mu = (0,0,1,0). \tag{37}$$

Then

$$W\varepsilon_1 = \begin{pmatrix} \alpha \\ 1 \\ 0 \\ \alpha \end{pmatrix} = \varepsilon_1 + \frac{\alpha}{E}k, \quad W\varepsilon_2 = \begin{pmatrix} \beta \\ 0 \\ 1 \\ \beta \end{pmatrix} = \varepsilon_2 + \frac{\beta}{E}k. \tag{38}$$

The shifted polarization remains transverse and has the same norm:

$$k \cdot (\varepsilon + ck) = k \cdot \varepsilon = 0, \quad (\varepsilon + ck)^2 = \varepsilon^2, \tag{39}$$

because $k^2 = 0$. We impose that the polarization vectors

$$\varepsilon^\mu \to \varepsilon^\mu + ck^\mu \tag{40}$$

must describe the same physical state.

## 1.2 Two-component Spinors

YL: [The lecture in this section is chaotic, if you feel unconfortable with the logic flow, it is totally normal. I tried my best to make it in a smooth way.]

### 1.2.1 Spinor Representations

We have discussed the representation of Lorentz group on the Hilbert space. Now we focus on the representation of Lorentz group on the space of fields. Considering the standard model, we mainly care about the spin 1/2 representations. The six Lorentz generators

$$J^i := \frac{1}{2}\varepsilon^{ijk}J^{jk}, \quad K^i := J^{0i} \tag{41}$$

satisfy

$$\begin{aligned} [J^i, J^j] &= i\varepsilon^{ijk}J^k, \\ [J^i, K^j] &= i\varepsilon^{ijk}K^k, \\ [K^i, K^j] &= -i\varepsilon^{ijk}J^k. \end{aligned} \tag{42}$$

Define the complex combinations

$$N_i^+ := \frac{1}{2}(J^i + iK^i), \quad N_i^- := \frac{1}{2}(J^i - iK^i). \tag{43}$$

They obey

$$\begin{aligned} \left[N_i^+, N_j^+\right] &= i\varepsilon_{ijk}N_k^+, \\ \left[N_i^-, N_j^-\right] &= i\varepsilon_{ijk}N_k^-, \\ \left[N_i^+, N_j^-\right] &= 0. \end{aligned} \tag{44}$$

Thus the complexified Lorentz algebra is two commuting copies of $\mathrm{su}(2)$, and finite-dimensional irreducible representations are labeled by $(j_L, j_R)$.

- For the undotted $\left(\frac{1}{2}, 0\right)$ representation,

$$J^i = \frac{\sigma^i}{2}, \quad K^i = -i\frac{\sigma^i}{2}. \tag{45}$$

A rotation by $\boldsymbol{\theta}$ and boost of rapidity $\boldsymbol{\eta}$ are represented by

$$\Lambda = \exp\left(i(\boldsymbol{\theta} - i\boldsymbol{\eta}) \cdot \frac{\boldsymbol{\sigma}}{2}\right) \in \mathrm{SL}(2, \mathbb{C}), \quad \det \Lambda = 1. \tag{46}$$

An undotted spinor transforms as

$$\lambda_\alpha \to \Lambda_\alpha{}^\beta \lambda_\beta. \tag{47}$$

> **Remark**
>
> By convention the LT matrix have the first index as the row index as lower index and the second index as the column index as upper index. Always stick to this convention.

- For the dotted $\left(0, \frac{1}{2}\right)$ representation,

$$J^i = \frac{\sigma^i}{2}, \quad K^i = +i\frac{\sigma^i}{2}, \tag{48}$$

and the conjugate spinor transforms as

$$\tilde{\lambda}_{\dot{\alpha}} \to (\Lambda^*)_{\dot{\alpha}}{}^{\dot{\beta}} \tilde{\lambda}_{\dot{\beta}} = \tilde{\lambda}_{\dot{\beta}} \left(\Lambda^\dagger\right)^{\dot{\beta}}{}_{\dot{\alpha}} \tag{49}$$

> **Remark**
>
> One important observation is that the complex conjugate of a (1/2, 0) representation is a (0, 1/2) representation. Thus, the dot and undotted indices can be viewed as related to each other with a CC.
>
> $$\lambda_\alpha^* = \tilde{\lambda}_{\dot{\alpha}}, \tag{50}$$
>
> The lorentz transformation of the (0,1/2) representation can also be written as the complex conjugate of the (1/2,0) representation as above:
>
> $$\tilde{\lambda}_{\dot{\alpha}} \to (\Lambda^*)_{\dot{\alpha}}{}^{\dot{\beta}} \tilde{\lambda}_{\dot{\beta}} \tag{51}$$
>
> Here we also take the convention that the first index is the row index and the second index is the column index, we only rename them as dotted indices to show that they live in the (0,1/2) representation space.

Now we have the basic conventions for the spinors and spinor indices. We can further define raising and lowering of spinor indices to define LT invariant contractions. Dotted and undotted indices are raised and lowered with antisymmetric tensors:

$$\lambda^\alpha = \varepsilon^{\alpha\beta}\lambda_\beta, \quad \lambda_\alpha = \varepsilon_{\alpha\beta}\lambda^\beta, \quad \varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta^\alpha_\gamma, \quad \varepsilon^{12} = 1, \quad \varepsilon_{12} = -1. \tag{52}$$

$$\tilde{\lambda}^{\dot{\alpha}} = \varepsilon^{\dot{\alpha}\dot{\beta}}\tilde{\lambda}_{\dot{\beta}}, \quad \tilde{\lambda}_{\dot{\alpha}} = \varepsilon_{\dot{\alpha}\dot{\beta}}\tilde{\lambda}^{\dot{\beta}}, \quad \varepsilon^{\dot{\alpha}\dot{\beta}}\varepsilon_{\dot{\beta}\dot{\gamma}} = \delta^{\dot{\alpha}}_{\dot{\gamma}}, \quad \varepsilon^{\dot{1}\dot{2}} = 1, \quad \varepsilon_{\dot{1}\dot{2}} = -1. \tag{53}$$

Under a Lorentz transformation, the four types of spinors transform as

$$\lambda_\alpha \to \Lambda_\alpha{}^\beta \lambda_\beta, \quad \lambda^\alpha \to \left(\Lambda^{-1}\right)_\beta{}^\alpha \lambda^\beta, \quad \tilde{\lambda}_{\dot{\alpha}} \to (\Lambda^*)_{\dot{\alpha}}{}^{\dot{\beta}} \tilde{\lambda}_{\dot{\beta}}, \quad \tilde{\lambda}^{\dot{\alpha}} \to \left((\Lambda^*)^{-1}\right)_{\dot{\beta}}{}^{\dot{\alpha}} \tilde{\lambda}^{\dot{\beta}}. \tag{54}$$

> **Remark**
>
> To consist with the following conventions, it is better to use $\Lambda^\dagger$ instead of $\Lambda^*$. (just take a transpose) Thus we can write the transformations of the dotted spinors as:
>
> $$\tilde{\lambda}_{\dot{\alpha}} \to \tilde{\lambda}_{\dot{\beta}} \Lambda^{\dagger\dot{\beta}}{}_{\dot{\alpha}}, \quad \tilde{\lambda}^{\dot{\alpha}} \to \left(\Lambda^\dagger\right)^{-1\dot{\alpha}}{}_{\dot{\beta}} \tilde{\lambda}^{\dot{\beta}}. \tag{55}$$
>
> now the upper index dotted spinor behaves like a ket.

We can thus see that the upper and lower indices transform in the opposite way, and the dotted and undotted indices transform in the complex conjugate way. The inverse property may give

us a chance of defining a Lorentz invariant contraction of spinors. We first introduce a braket notation for the spinors, such that the contraction of vectors gives a LT invariant scalar. The spinor-helicity bra-ket notation is

$$\langle\lambda| \coloneqq \lambda^\alpha, \quad |\lambda\rangle \coloneqq \lambda_\alpha, \quad [\lambda| \coloneqq \tilde{\lambda}_{\dot{\alpha}}, \quad |\lambda] \coloneqq \tilde{\lambda}^{\dot{\alpha}}. \tag{56}$$

The two Lorentz-invariant contractions are

$$\langle pq\rangle \coloneqq p^\alpha q_\alpha = \varepsilon^{\alpha\beta} p_\beta q_\alpha, \tag{57}$$

$$[pq] \coloneqq \tilde{p}_{\dot{\alpha}} \tilde{q}^{\dot{\alpha}}. \tag{58}$$

> **Remark**
>
> One should be careful about the ordering of the contraction. In the $\langle\rangle$ contraction, the first one have upper index and the second one have lower index. In the [] contraction, the first one have lower index and the second one have upper index. This is in general important due to the fact that lower and upper $\varepsilon$ are not the same and both antisymmetric.

Both are antisymmetric:

$$\langle pq\rangle = -\langle qp\rangle, \quad [pq] = -[qp], \quad \langle pp\rangle = [pp] = 0. \tag{59}$$

### 1.2.2 Momentum as a bispinor

Indeed the spin 1/2 field can be taken as a 2 component spinor field. Moreover, we can also represent 4 vectors as composite spinors. A four-momentum can be mapped to a $2 \times 2$ matrix:

$$p_{\alpha\dot{\alpha}} \coloneqq p_\mu \sigma^\mu_{\alpha\dot{\alpha}} = \begin{pmatrix} p^0 - p^3 & -p^1 + ip^2 \\ -p^1 - ip^2 & p^0 + p^3 \end{pmatrix}. \tag{60}$$

Defining

$$p^+ \coloneqq p^0 + p^3, \quad p^- \coloneqq p^0 - p^3, \quad p_\perp \coloneqq p^1 + ip^2, \tag{61}$$

we may also write

$$p_{\alpha\dot{\alpha}} = \begin{pmatrix} p^- & -p_\perp^* \\ -p_\perp & p^+ \end{pmatrix}. \tag{62}$$

If we do a Lorentz transformation to the 4 momentum, the bispinor transforms as the spinors with suitable index dot and position: (that's why this is called a bispinor not meerly a matrix)

$$p_{\alpha\dot{\alpha}} \to \Lambda_\alpha{}^\beta p_{\beta\dot{\beta}} \Lambda^{\dagger\,\dot{\beta}}{}_{\dot{\alpha}}. \tag{63}$$

> **Remark**
>
> Notice that the dotted and undotted and upper and lower indices are defined to suite the convention of lorentz transformation. The index always transforms as 2 component spinors with the same position and doted or undotted.

Its determinant is the Lorentz invariant

$$\det p_{\alpha\dot{\alpha}} = (p^0)^2 - (p^1)^2 - (p^2)^2 - (p^3)^2 = p^2 = m^2. \tag{64}$$

Therefore

- for a massive momentum, $\det p \neq 0$ and $\mathrm{rank}(p) = 2$;

$$p^2 \neq 0: \quad \det(p_{\alpha\dot{\alpha}}) \neq 0, \quad \operatorname{rank}(p_{\alpha\dot{\alpha}}) = 2, \tag{65}$$

- for a massless momentum, $\det p = 0$ and $\operatorname{rank}(p) = 1$.

$$p^2 = 0: \quad \det(p_{\alpha\dot{\alpha}}) = 0, \quad \operatorname{rank}(p_{\alpha\dot{\alpha}}) = 1. \tag{66}$$

> **Remark**
>
> By convention, we write $\sigma^{\mu}{}_{\alpha\dot{\alpha}}$ and $\overline{\sigma}^{\mu\dot{\alpha}\alpha}$. They are related by
>
> $$\overline{\sigma}^{\mu\dot{\alpha}\alpha} = \varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}\sigma^{\mu}{}_{\beta\dot{\beta}}. \tag{67}$$
>
> When raising or lowering both indices, their order is exchanged. This is to suit the convention of the matrix and the first index is always the row index and the second is the column index. Thus the momentum matrix under raising of indices is:
>
> $$p^{\dot{\alpha}\alpha} = p_{\mu}\overline{\sigma}^{\mu\dot{\alpha}\alpha} = \begin{pmatrix} p^0 + p^3 & p^1 - ip^2 \\ p^1 + ip^2 & p^0 - p^3 \end{pmatrix}. \tag{68}$$

## 1.3 Spinors for Massless Particles

### 1.3.1 Momentum and Basis Spinors

In the massless case, the rank-one matrix factorizes:

$$p_{\alpha\dot{\alpha}} = \lambda_{\alpha}\tilde{\lambda}_{\dot{\alpha}} = |p\rangle[p|. \tag{69}$$

Choose the reference momentum

$$k^{\mu} = E(1, 0, 0, 1). \tag{70}$$

With the convention $k_{\alpha\dot{\alpha}} = k_{\mu}\sigma^{\mu}_{\alpha\dot{\alpha}}$ and $k^{\dot{\alpha}\alpha} = k_{\mu}\overline{\sigma}^{\mu\dot{\alpha}\alpha}$, we have:

$$k^{\dot{\alpha}\alpha} = 2E\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = k^{\dot{\alpha}}k^{\alpha}. \tag{71}$$

Under a Lorentz transformation, the dot on the index and the position of the index

$$p^{\dot{\alpha}\alpha} = \Lambda^{\dagger}{}^{\dot{\alpha}}{}_{\dot{\beta}}k^{\dot{\beta}\beta}\Lambda_{\beta}{}^{\alpha} = p^{\dot{\alpha}}p^{\alpha}. \tag{72}$$

For a real null momentum of energy $E$ and direction $(\theta, \varphi)$:

$$p^{\mu} = E(1, \sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta), \tag{73}$$

one may choose the following spinors to represent the momentum:

$$|p\rangle = \sqrt{2E}\begin{pmatrix} -e^{-i\varphi}\sin\left(\frac{\theta}{2}\right) \\ \cos\left(\frac{\theta}{2}\right) \end{pmatrix}, \quad |p] = \sqrt{2E}\begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ e^{i\varphi}\sin\left(\frac{\theta}{2}\right) \end{pmatrix}. \tag{74}$$

remember the difference between bras and kets are raising and lowering index and the difference between $|p\rangle$ and $|p]$ is complex conjugation and a raising and lowering of index. The remaining spinors are obtained by raising and lowering indices:

$$p_{\dot{\alpha}} = \varepsilon_{\dot{\alpha}\dot{\beta}}p^{\dot{\beta}}, \quad p^{\alpha} = (p^{\dot{\alpha}})^*, \quad p_{\alpha} = \varepsilon_{\alpha\beta}p^{\beta}. \tag{75}$$

Though the momentum is invariant under little group transformations, the spinor that constructs the momentum should be not. And it will in fact transform covariantly under the little

group by the definition of spinor transformation. A little-group rotation around the direction of $p$ acts as

$$|p\rangle \to e^{i\frac{\phi}{2}}|p\rangle, \quad |p] \to e^{-i\frac{\phi}{2}}|p], \tag{76}$$

and leaves $p_{\alpha\dot{\alpha}} = |p\rangle[p|$ invariant. Thus, everything is consistent. We say that the spinor $|p\rangle$ has helicity $-\frac{1}{2}$ and the spinor $|p]$ has helicity $+\frac{1}{2}$, from now on we will use helicity weight to denote how a quantity transform under rotation around a certain direction (most probabaly the momentum direction).

> **Remark**
>
> For a real null momentum, $|p\rangle$ and $|p]$ are defined only up to opposite phase factors:
>
> $$|p\rangle \to e^{i\phi}|p\rangle, \quad |p] \to e^{-i\phi}|p]. \tag{77}$$
>
> one may see that this corresponds to the little group transformation of rotation around the momentum direction.

One now may realize that the spinors constructed here can natually be the wave function of spin 1/2 particles (i.e. when Weyl fermion act on vaccum gives a spinor wave function) with fixed chiralities and helicities $u_+, u_-$.The result is that:

> **Remark**
>
> Here we use the 4 component spinors notation for convenience in calculation. Though it can understand as 2 Weyl spinors combined together.

$$\begin{aligned} u_-(p) = |p\rangle, \quad u_+(p) = |p], \quad v_+(p) = |p\rangle, \quad v_-(p) = |p], \\ \overline{u}_-(p) = [p|, \quad \overline{u}_+(p) = \langle p|, \quad \overline{v}_+(p) = [p|, \quad \overline{v}_-(p) = \langle p|. \end{aligned} \tag{78}$$

Here we abuse the notation to use 2 component spinors to represent 4 component spinors. What we really mean is that:

$$u_-(p) = \begin{pmatrix} |p\rangle \\ 0 \end{pmatrix}, \quad u_+(p) = \begin{pmatrix} 0 \\ |p] \end{pmatrix}, \tag{79}$$

$$v_+(p) = \begin{pmatrix} |p\rangle \\ 0 \end{pmatrix}, \quad v_-(p) = \begin{pmatrix} 0 \\ |p] \end{pmatrix}. \tag{80}$$

$$\overline{u}_-(p) = (0, [p|), \quad \overline{u}_+(p) = (\langle p|, 0), \tag{81}$$

$$\overline{v}_+(p) = (0, [p|), \quad \overline{v}_-(p) = (\langle p|, 0). \tag{82}$$

The $|\ p\rangle$ ket always stays on the upper of left of the 4 component spinor and the $|\ p]$ ket always stays on the lower or right of the 4 component spinor. Another commonly used notation is that:

$$p_\mu \gamma^\mu = |p\rangle[p| + |p]\langle p|. \tag{83}$$

All this series of notation makes it convenience to calculate diagrams containing fermions.

### 1.3.2 Polarization Vectors

Now we turn to the polarization vectors of the massless vector fields. We know that under Wigner's little group the polarization vectors transform nontrivially. Thus it need more than the momentum spinors to construct the polarization vectors. We introduce an auxiliary null reference momentum $\xi$ such that

$$\xi^2 = 0, \quad \langle p\xi\rangle \neq 0, \quad [p\xi] \neq 0. \tag{84}$$

$$\varepsilon^+_{\alpha\dot\alpha}(p,\xi) = \sqrt{2}\frac{1}{\langle \xi p\rangle}|\xi\rangle[p|, \tag{85}$$

$$\varepsilon^-_{\alpha\dot\alpha}(p,\xi) = \sqrt{2}\frac{1}{[p\xi]}(|p\rangle[\xi|). \tag{86}$$

One can see that they satisfy

$$p\cdot\varepsilon^\pm = 0, \quad \varepsilon^+\cdot\varepsilon^- = -1. \tag{87}$$

Moreover, under little group rotations they transform nontrivially:

$$\varepsilon^\pm \to e^{\pm i\phi}\varepsilon^\pm. \tag{88}$$

This is consistent with the fact that the polarization vectors are transverse to the momentum and have helicity $\pm 1$.

The reference spinor is arbitrary. Since $\langle p\xi\rangle \neq 0$, another choice can be written as

$$|\xi'\rangle = a|\xi\rangle + b|p\rangle. \tag{89}$$

Then

$$\varepsilon^+_{\alpha\dot\alpha}(p,\xi') = \varepsilon^+_{\alpha\dot\alpha}(p,\xi) + \frac{\sqrt{2}b}{a\langle\xi p\rangle}p_{\alpha\dot\alpha}. \tag{90}$$

Thus changing the reference spinor shifts the polarization by a term proportional to the momentum. The requirement that observables be independent of $\xi$ is the gauge redundancy.

For sake of calculation, we have another expression of the polarization vectors in terms of the 4 compnent spinors and the gamma matrices:

$$\varepsilon^\mu_-(p;q) = \frac{\langle p|\gamma^\mu|q]}{\sqrt{2}[qp]}, \quad \varepsilon^\mu_+(p;q) = \frac{\langle q|\gamma^\mu|p]}{\sqrt{2}\langle qp\rangle}. \tag{91}$$

## 1.4 Spinors for massive particles

### 1.4.1 Momentum and Basis Spinors

For a massive momentum,

$$\det(p_{\alpha\dot\alpha}) = p^2 = m^2 \neq 0, \tag{92}$$

so the rank-two bispinor can be decomposed into two rank-one matrices:

$$p_{\alpha\dot\alpha} = l^1_{\alpha\dot\alpha} + l^2_{\alpha\dot\alpha} = |1\rangle[1| + |2\rangle[2|, \quad \left(l^1\right)^2 = \left(l^2\right)^2 = 0. \tag{93}$$

This decomposition is not unique. In the rest frame,

$$k^\mu = (m,0,0,0), \quad l^\mu_1 = \frac{m}{2}(1,\boldsymbol{n}), \quad l^\mu_2 = \frac{m}{2}(1,-\boldsymbol{n}). \tag{94}$$

The two null rays define the spin-quantization axis (just a stupid name for now)

$$n^\mu = \frac{1}{m}(l^\mu_1 - l^\mu_2), \quad n^2 = -1, \quad n\cdot k = 0. \tag{95}$$

**Remark**

This axis is not unique, we have a freedom of choosing an internal direction. One can understand as we use two massless particles moving in opposite directions to construct a massive particle momentum.

Under a rotation by an angle $\phi$ around $\boldsymbol{n}$, the two null momenta $l_1$ and $l_2$ remain invariant. Using the spinor transformation rules, we find

$$|1] \to e^{-i\frac{\phi}{2}}|1], \quad |1\rangle \to e^{i\frac{\phi}{2}}|1\rangle, \quad |2] \to e^{i\frac{\phi}{2}}\,|2], \quad |2\rangle \to e^{-i\frac{\phi}{2}}|2\rangle. \tag{96}$$

Thus we say that the spinors have the "helicity" of:

- $|1]$: $h = +\frac{1}{2}$ $|1\rangle$: $h = -\frac{1}{2}$ $|2]$: $h = -\frac{1}{2}$ $|2\rangle$: $h = +\frac{1}{2}$

We may notice that the helicity of 1 and 2 are opposite, this is due to they in fact are moving in opposite directions. We take the direction of $l_1$ as the spin-quantization axis, thus the spinors associated with $l_2$ have opposite helicity.

**Remark**

Notice that this is not true helicity, cause it is just the "spin" along a chosen axis. But we split the massless momentum using this axis as the momentum direction, thus we call it helicity.

Before we notice that the massless momentum spinors naturally give us the "wavefunction" $u_+, u_-$ (sorry for not having a better phrase) of the Weyl fermions. Now we can also see that the massive momentum spinors naturally give us the "wavefunction" of the Dirac fermions with spin quantized around $n$ direction. They are:

$$u_+(k) = \begin{pmatrix} -|2\rangle \\ |1] \end{pmatrix}, \quad u_-(k) = \begin{pmatrix} |1\rangle \\ |2] \end{pmatrix}. \tag{97}$$

**Remark**

Quantized around $n$ direction means that the spinor we take is eigenstate of the rotational generator along $n$ direction of the spinor representation of lorentz group.

Taking $n$ direction as $\boldsymbol{n} = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$, we have the explicit from (up to a relative phase factor):

$$u_+(k) = \sqrt{m}\begin{pmatrix} \cos\left(\frac{\theta}{2}\right) \\ e^{i\varphi}\sin\left(\frac{\theta}{2}\right) \\ \cos\left(\frac{\theta}{2}\right) \\ e^{i\varphi}\sin\left(\frac{\theta}{2}\right) \end{pmatrix}, \quad u_-(k) = \sqrt{m}\begin{pmatrix} -e^{-i\varphi}\sin\left(\frac{\theta}{2}\right) \\ \cos\left(\frac{\theta}{2}\right) \\ -e^{-i\varphi}\sin\left(\frac{\theta}{2}\right) \\ \cos\left(\frac{\theta}{2}\right) \end{pmatrix}. \tag{98}$$

One can prove that they solve the Dirac equation in the rest frame. Moreover, the commonly used basis for dirac spinors which is:

$$u_+(k) = \sqrt{m}\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_-(k) = \sqrt{m}\begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \tag{99}$$

is just the special case of $\boldsymbol{n} = (0, 0, 1)$.

In the rest frame,

$$k_{\alpha\dot{\alpha}} = m\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}. \tag{100}$$

A rotation leaves $k$ invariant, however it rotates the two null momenta $l_1$ and $l_2$:

$$RkR^\dagger = k, \quad Rl^iR^\dagger \neq l^i. \tag{101}$$

Thus the massive little group acts nontrivially on the two null momenta and rotates the spin-quantization axis.

For a general momentum, we can decompose it into the following two parts:

$$p^\mu = l^\mu + \frac{m^2}{2l\cdot q}q^\mu, \quad l^2 = q^2 = 0. \tag{102}$$

We can find that $p^2 = m^2$. A convenient spinor decomposition is

$$|1\rangle = |l\rangle, \quad |2\rangle = \frac{m}{\langle ql\rangle}|q\rangle, \quad |1] = |l], \quad |2] = \frac{m}{[lq]}|q], \tag{103}$$

where we use a useful identity is for any null $l$ and $q$,

$$2l\cdot q = \langle lq\rangle[ql]. \tag{104}$$

Moreover, we can also combine the two spinor together and amazingly find a way to solve the dirac equation for a general momentum and get the wavefunctions of the Dirac fermions:

$$u_+(p) = \begin{pmatrix}\frac{m}{\langle lq\rangle}|q\rangle\\ |l]\end{pmatrix}, \quad u_-(p) = \begin{pmatrix}|l\rangle\\ \frac{m}{[lq]}|q]\end{pmatrix}. \tag{105}$$

We know that the Basis of "wavefunction" is not unique and depend on the rest frame spinor basis choice and a boost choice. This choice of $l, q$ decomposition implicitly chooses a basis.

Now we consider the massless limit. For any fixed $q$,

$$m^2 \to 0 \quad \Rightarrow \quad p^\mu \to l^\mu. \tag{106}$$

The wavefunctions reduce to

$$u_+(p) = \begin{pmatrix}\frac{m}{\langle lq\rangle}|q\rangle\\ |l]\end{pmatrix} \to \begin{pmatrix}0\\ |l]\end{pmatrix}, \quad u_-(p) = \begin{pmatrix}|l\rangle\\ \frac{m}{[lq]}|q]\end{pmatrix} \to \begin{pmatrix}|l\rangle\\ 0\end{pmatrix}. \tag{107}$$

This is a result we are familiar with which is massless Dirac spinor has a single nonvanishing chirality.

### 1.4.2 Polarization Vectors

The massive-vector polarization bispinors are

$$\varepsilon^+ = \frac{|1]\langle 2|}{\sqrt{2}m}, \quad \varepsilon^- = \frac{|2]\langle 1|}{\sqrt{2}m}, \quad \varepsilon^0 = \frac{|1]\langle 1| - |2]\langle 2|}{m}. \tag{108}$$

They have the correct little-group transformation and obey

$$p\cdot\varepsilon = 0. \tag{109}$$

We now take the massless limit. transverse polarizations remain finite:

$$\varepsilon^+ \to \frac{|\,l]\langle q\,|}{\sqrt{2}\langle lq\rangle} \quad \varepsilon^- \to \frac{|\,l\rangle[q\,|}{\sqrt{2}[lq]}. \tag{110}$$

We can see $q$ now becomes the massless reference vector. A normalized longitudinal polarization can be written as

$$\varepsilon_0^\mu(p;q) = \frac{1}{m}\left(l^\mu - \frac{m^2}{2l\cdot q}q^\mu\right) = \frac{p^\mu}{m} - \frac{m}{p\cdot q}q^\mu. \tag{111}$$

Its massless limit is singular, but the divergent part is proportional to $p^\mu$ and is therefore pure gauge.

### 1.4.3 From polarizations to fields

In QFT, we play with fields. The vector field can be expanded in the following terms in terms of the polarization vector.

$$A_\mu(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2E_p}\sum_{h=\pm1}\left(\varepsilon_\mu^h(p)a_h(p)e^{-ip\cdot x} + \varepsilon_\mu^{h*}(p)a_h^\dagger(p)e^{ip\cdot x}\right). \tag{112}$$

For

$$|(p,h)\rangle = a_h^\dagger(p)|0\rangle, \tag{113}$$

the field-state overlap is

$$\langle 0|A_\mu(x)|(p,h)\rangle = \varepsilon_\mu^h(p)e^{-ip\cdot x}. \tag{114}$$

If the polarization is shifted by

$$\varepsilon_\mu^h(p) \to \varepsilon_\mu^h(p) + c(p)p_\mu, \tag{115}$$

then, after absorbing an overall factor of $i$ into $\alpha$,

$$\delta A_\mu(x) = \int \frac{\mathrm{d}^3p}{(2\pi)^3 2E_p}c(p)p_\mu a_h(p)e^{-ip\cdot x} = \partial_\mu\alpha(x). \tag{116}$$

Hence a change of reference spinor is equivalent to the gauge transformation

$$A_\mu \to A_\mu + \partial_\mu\alpha. \tag{117}$$

# 2 QED and Non-Abelian Gauge Theories

## 2.1 Quantum electrodynamics

### 2.1.1 Vector Field

A relativistic quantum field theory containing a massless spin-one particle can be described using a vector field $A_\mu$ with the redundancy

$$A_\mu(x) \to A_\mu(x) + \partial_\mu \alpha(x), \quad \alpha(x) \in \mathbb{R}. \tag{118}$$

The basic gauge-invariant local object is the field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad F_{\mu\nu} = -F_{\nu\mu}. \tag{119}$$

The unique parity-even gauge-invariant kinetic term with two derivatives is

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}. \tag{120}$$

In four spacetime dimensions,

$$\left[A_\mu\right] = 1, \quad \left[F_{\mu\nu}\right] = 2, \quad [\psi] = \frac{3}{2}. \tag{121}$$

A possible dimension-six contraction of three field strengths vanishes:

$$F_\mu{}^\nu F_\nu{}^\rho F_\rho{}^\mu = 0. \tag{122}$$

The leading pure-photon corrections therefore occur at dimension eight. For example,

$$\Delta\mathcal{L} = \frac{c_1}{\Lambda^4}\left(F_{\mu\nu}F^{\mu\nu}\right)^2 + \frac{c_2}{\Lambda^4}\left(F_{\mu\nu}\tilde{F}^{\mu\nu}\right)^2 + ..., \tag{123}$$

where

$$\tilde{F}^{\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}. \tag{124}$$

Operators of dimension greater than four are irrelevant in the infrared. Restricting temporarily to operators of dimension at most four gives the free Maxwell theory. Such a theory is conventionally called renormalizable, although higher-dimensional operators are perfectly meaningful within effective field theory.

### 2.1.2 Coupling photons to matter

Electrons are described by a spin-$\frac{1}{2}$ Dirac field. A manifestly gauge-invariant photon-matter operator is the dipole interaction

$$\Delta\mathcal{L}_{\text{dipole}} = \frac{c}{\Lambda}\overline{\psi}\sigma^{\mu\nu}\psi F_{\mu\nu}, \tag{125}$$

with

$$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]. \tag{126}$$

This operator has dimension five and is therefore irrelevant. The renormalizable interaction instead couples the vector potential directly to a current:

$$S_{\text{int}} = -\int \mathrm{d}^4x A_\mu(x) J^\mu(x). \tag{127}$$

Under

$$\delta A_\mu = \partial_\mu \alpha, \tag{128}$$

the interaction changes by

$$\delta S_{\text{int}} = -\int \mathrm{d}^4x \big(\partial_\mu \alpha\big) J^\mu = \int \mathrm{d}^4x \alpha \partial_\mu J^\mu, \tag{129}$$

where a boundary term was discarded. Gauge invariance therefore requires

$$\partial_\mu J^\mu = 0. \tag{130}$$

This conservation law is automatic when $A_\mu$ enters only through the field strength. Indeed, for an antisymmetric tensor $O^{\mu\nu}$,

$$F_{\mu\nu} O^{\mu\nu} = 2\big(\partial_\mu A_\nu\big) O^{\mu\nu} \to -2 A_\nu \partial_\mu O^{\mu\nu}, \tag{131}$$

where the arrow means equality inside the action up to a boundary term. The corresponding current is

$$J^\nu = 2\partial_\mu O^{\mu\nu}, \tag{132}$$

and hence

$$\partial_\nu J^\nu = 2\partial_\nu \partial_\mu O^{\mu\nu} = 0 \tag{133}$$

identically. When $A_\mu$ appears directly in the action, however, current conservation becomes a genuine constraint on the matter theory.

The free Dirac Lagrangian is

$$\mathcal{L}_{\text{matter}} = i\overline{\psi}\gamma^\mu \partial_\mu \psi - m\overline{\psi}\psi. \tag{134}$$

It is invariant under the global transformation

$$\psi(x) \to e^{-ie\alpha}\psi(x), \quad \overline{\psi}(x) \to \overline{\psi}(x) e^{ie\alpha}, \tag{135}$$

The associated Noether current is

$$j^\mu = -i\frac{\partial \mathcal{L}_{\text{matter}}}{\partial \partial_\mu \psi}\psi = \overline{\psi}\gamma^\mu \psi, \tag{136}$$

Coupling this current to the photon field gives the interaction term and if we promote the symmetry to local one, then it is invariant under gauge transformations.

We use the mutually consistent transformations

$$A_\mu(x) \to A_\mu(x) + \partial_\mu \alpha(x), \tag{137}$$

$$\psi(x) \to \psi'(x) = e^{-ie\alpha(x)}\psi(x), \tag{138}$$

$$\overline{\psi}(x) \to \overline{\psi}'(x) = \overline{\psi}(x) e^{ie\alpha(x)}. \tag{139}$$

An ordinary derivative does not transform covariantly. Define the covariant derivative by

$$D_\mu = \partial_\mu + ieA_\mu. \tag{140}$$

Under the joint transformation,

$$D'_\mu \psi'(x) = \big(\partial_\mu + ie\big(A_\mu + \partial_\mu \alpha\big)\big) e^{-ie\alpha}\psi. \tag{141}$$

Expanding the derivative gives

$$D'_\mu \psi'(x) = e^{-ie\alpha}\big(\partial_\mu - ie\partial_\mu\alpha + ieA_\mu + ie\partial_\mu\alpha\big)\psi \tag{142}$$

$$= e^{-ie\alpha} D_\mu \psi. \tag{143}$$

Thus $D_\mu\psi$ transforms in exactly the same way as $\psi$.

### 2.1.3 The QED Lagrangian

The gauge-invariant QED Lagrangian is

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\overline{\psi}\gamma^\mu D_\mu\psi - m\overline{\psi}\psi. \tag{144}$$

Expanding the covariant derivative gives

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\overline{\psi}\gamma^\mu\partial_\mu\psi - m\overline{\psi}\psi - eA_\mu\overline{\psi}\gamma^\mu\psi. \tag{145}$$

- This is an Abelian gauge theory with gauge group $U(1)$.
- Gauge invariance forbids a photon mass term

$$\frac{1}{2}m_A^2 A_\mu A^\mu. \tag{146}$$

- The replacement

$$\partial_\mu \to D_\mu \tag{147}$$

  produces the minimal photon-matter interaction.
- The Abelian field strength is linear in $A_\mu$, so the renormalizable Maxwell term contains no photon self-interactions.

Varying the action with respect to $A_\mu$ gives

$$\partial_\nu F^{\nu\mu} = J^\mu, \quad J^\mu = e\overline{\psi}\gamma^\mu\psi. \tag{148}$$

These are the inhomogeneous Maxwell equations. Taking the divergence implies current conservation:

$$\partial_\mu J^\mu = \partial_\mu\partial_\nu F^{\nu\mu} = 0. \tag{149}$$

The homogeneous Maxwell equations follow from the Bianchi identity

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0, \tag{150}$$

or equivalently

$$\varepsilon^{\mu\nu\rho\sigma}\partial_\nu F_{\rho\sigma} = 0. \tag{151}$$

Finally, the interaction term

$$\mathcal{L}_{\text{int}} = -eA_\mu\overline{\psi}\gamma^\mu\psi \tag{152}$$

gives the electron-photon vertex

$$-ie\gamma^\mu. \tag{153}$$

## 2.2 Non-Abelian gauge theories

### 2.2.1 Motivation

When QED is restricted to operators of dimension at most four, the photon has no self-interactions. Yang-Mills theory generalizes the Abelian gauge principle and describes interacting massless spin-one fields.

The simplest example is based on the group SU(2). Let the matter field transform as a doublet,

$$\psi(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix}. \tag{154}$$

The generators in the fundamental representation are

$$T^a = \frac{\sigma^a}{2}, \quad [T^a, T^b] = i\varepsilon^{abc}T^c, \quad \mathrm{tr}\big(T^aT^b\big) = \frac{1}{2}\delta^{ab}. \tag{155}$$

For a general compact Lie group $G$, we write

$$[T^a, T^b] = if^{abc}T^c. \tag{156}$$

The structure constants of SU(2) are

$$f^{abc} = \varepsilon^{abc}. \tag{157}$$

### 2.2.2 From a global symmetry to a local symmetry

The free matter Lagrangian is

$$\mathcal{L}_0 = i\overline{\psi}\gamma^\mu\partial_\mu\psi - m\overline{\psi}\psi. \tag{158}$$

It is invariant under a global transformation

$$\psi(x) \to \psi'(x) = U\psi(x), \tag{159}$$

$$\overline{\psi}(x) \to \overline{\psi}'(x) = \overline{\psi}(x)U^{-1}, \tag{160}$$

where

$$U = e^{-ig\alpha^aT^a}, \quad \partial_\mu U = 0. \tag{161}$$

If $U = U(x)$ is local, the ordinary derivative produces an extra term:

$$\overline{\psi}'\gamma^\mu\partial_\mu\psi' = \overline{\psi}\gamma^\mu\partial_\mu\psi + \overline{\psi}\gamma^\mu U^{-1}\big(\partial_\mu U\big)\psi. \tag{162}$$

To compensate for this term, introduce a Lie-algebra-valued gauge field

$$A_\mu \equiv A_\mu{}^aT^a \tag{163}$$

and define the covariant derivative

$$D_\mu = \partial_\mu - igA_\mu = \partial_\mu - igA_\mu{}^aT^a. \tag{164}$$

We demand that it transform covariantly:

$$D'_\mu\psi'(x) = U(x)D_\mu\psi(x). \tag{165}$$

### 2.2.3 Gauge-field transformation

Using $\psi' = U\psi$, covariance requires

$$(\partial_\mu - igA'_\mu)U\psi = U(\partial_\mu - igA_\mu)\psi. \tag{166}$$

Therefore,

$$A'_\mu = UA_\mu U^{-1} - \frac{i}{g}(\partial_\mu U)U^{-1}. \tag{167}$$

This is the finite transformation law of a non-Abelian connection. It contains both a homogeneous rotation and an inhomogeneous derivative term.

For an infinitesimal transformation,

$$U(x) = 1 - ig\alpha^a(x)T^a + O(\alpha^2), \tag{168}$$

so that

$$A'_\mu{}^a = A_\mu{}^a - \partial_\mu\alpha^a + gf^{abc}\alpha^b A_\mu{}^c. \tag{169}$$

The homogeneous part shows that the gauge fields transform in the adjoint representation. Unlike the photon, the non-Abelian gauge bosons therefore carry gauge charge.

With the convention used here, the Abelian limit is

$$A_\mu \to A_\mu - \partial_\mu\alpha. \tag{170}$$

Replacing $\alpha \to -\alpha$ gives the alternative QED convention.

### 2.2.4 Field strength from the covariant derivative

The non-Abelian field strength is defined by

$$[D_\mu, D_\nu] = -igF_{\mu\nu}. \tag{171}$$

Using $D_\mu = \partial_\mu - igA_\mu$ gives

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]. \tag{172}$$

In components,

$$F_{\mu\nu}{}^a = \partial_\mu A_\nu{}^a - \partial_\nu A_\mu{}^a + gf^{abc}A_\mu{}^b A_\nu{}^c. \tag{173}$$

Because

$$D'_\mu = UD_\mu U^{-1}, \tag{174}$$

the field strength transforms homogeneously:

$$F'_{\mu\nu} = UF_{\mu\nu}U^{-1}. \tag{175}$$

Infinitesimally,

$$F'_{\mu\nu}{}^a = F_{\mu\nu}{}^a + gf^{abc}\alpha^b F_{\mu\nu}{}^c. \tag{176}$$

Thus the field strength transforms in the adjoint representation. It is gauge covariant rather than gauge invariant.

### 2.2.5 The Yang-Mills Lagrangian

The gauge-invariant kinetic term is obtained by tracing over the gauge indices:

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}). \tag{177}$$

For the normalization

$$\mathrm{tr}\big(T^a T^b\big) = \frac{1}{2}\delta^{ab}, \tag{178}$$

this becomes

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4} F_{\mu\nu}{}^a F^{\mu\nu a}. \tag{179}$$

Gauge invariance follows from cyclicity of the trace:

$$\mathrm{tr}\big(F'_{\mu\nu} F'^{\mu\nu}\big) = \mathrm{tr}\big(U F_{\mu\nu} U^{-1} U F^{\mu\nu} U^{-1}\big) \tag{180}$$

$$= \mathrm{tr}\big(F_{\mu\nu} F^{\mu\nu}\big). \tag{181}$$

Adding Dirac matter in a representation $R$ gives

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}{}^a F^{\mu\nu a} + i\bar{\psi}\gamma^\mu D_\mu \psi - m\bar{\psi}\psi, \tag{182}$$

with

$$D_\mu = \partial_\mu - igA_\mu{}^a T_R^a, \tag{183}$$

and

$$\big[T_R^a, T_R^b\big] = i f^{abc} T_R^c. \tag{184}$$

The same coupling $g$ controls both the coupling to matter and the self-interactions of the gauge fields. Gauge invariance does not allow these couplings to be chosen independently.

### 2.2.6 Gauge-boson self-interactions

Define the linearized field strength

$$f_{\mu\nu}{}^a = \partial_\mu A_\nu{}^a - \partial_\nu A_\mu{}^a. \tag{185}$$

Then

$$F_{\mu\nu}{}^a = f_{\mu\nu}{}^a + g f^{abc} A_\mu{}^b A_\nu{}^c. \tag{186}$$

Expanding the Yang-Mills kinetic term gives quadratic, cubic, and quartic terms:

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4} f_{\mu\nu}{}^a f^{\mu\nu a} - \frac{g}{2} f^{abc} f_{\mu\nu}{}^a A^{\mu b} A^{\nu c} - \frac{g^2}{4} f^{abc} f^{ade} A_\mu{}^b A_\nu{}^c A^{\mu d} A^{\nu e}. \tag{187}$$

The cubic and quartic terms describe three- and four-gauge-boson interactions. Since $f^{abc}$ is antisymmetric, there is no self-coupling involving only one gauge component, but distinct gauge components interact with one another.

### 2.2.7 General compact gauge groups

For a representation $R$, the invariant quadratic form is

$$\mathrm{tr}_R\big(T_R^a T_R^b\big) = T(R)\delta^{ab}. \tag{188}$$

A positive-definite gauge kinetic energy requires a positive invariant inner product on the Lie algebra. And the matter field we want a simple kinetic term. This naturally leads to compact gauge groups represented by unitary matrices,

$$U^\dagger U = 1. \tag{189}$$

Important compact possibilities include

- $\mathrm{SU}(N)$,

- $\mathrm{SO}(N)$,
- $\mathrm{Sp}(N)$,
- the exceptional groups $G_2$, $F_4$, $E_6$, $E_7$, and $E_8$,
- products of these groups together with Abelian $U(1)$ factors.

For a general group, the SU(2) replacements are

$$T^a \to T^a_R, \quad \varepsilon^{abc} \to f^{abc}. \tag{190}$$

### 2.2.8 Tree-level Feynman rules

The following rules use the convention

$$D_\mu = \partial_\mu - igA_\mu \tag{191}$$

The matter-gauge-boson vertex is

$$ig\gamma^{\mu\alpha}{}_\beta T^{a\,i}{}_j. \tag{192}$$

The gauge-boson propagator is

$$\frac{-i\eta^{\mu\nu}\delta^{ab}}{p^2 + i\varepsilon}. \tag{193}$$

The fermion propagator is

$$\frac{i\big(\gamma^\rho p_\rho + m\big)^\alpha{}_\beta}{p^2 - m^2 + i\varepsilon}. \tag{194}$$

For the three-gauge-boson vertex, take all momenta incoming and impose

$$p_1 + p_2 + p_3 = 0. \tag{195}$$

For external legs $(a,\mu)$, $(b,\nu)$, and $(c,\rho)$, the vertex is

$$V_3 = gf^{abc}[\eta^{\mu\nu}(p_2 - p_1)^\rho + \eta^{\nu\rho}(p_3 - p_2)^\mu + \eta^{\rho\mu}(p_1 - p_3)^\nu]. \tag{196}$$

For the four-gauge-boson vertex with external legs $(a,\mu)$, $(b,\nu)$, $(c,\rho)$, and $(d,\sigma)$, write

$$V_4 = -ig^2(C_1 + C_2 + C_3), \tag{197}$$

where

$$C_1 = f^{abe}f^{cde}(\eta^{\mu\rho}\eta^{\nu\sigma} - \eta^{\mu\sigma}\eta^{\nu\rho}), \tag{198}$$

$$C_2 = f^{ace}f^{bde}(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\sigma}\eta^{\nu\rho}), \tag{199}$$

$$C_3 = f^{ade}f^{bce}(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma}). \tag{200}$$

These rules are sufficient for tree-level calculations involving physical external polarizations. At loop level, a consistent gauge-fixed quantization also requires the Faddeev-Popov ghost sector.

# 3 Tree-level amplitudes in gauge theory

These notes use the all-incoming convention for each amplitude, all momenta are incoming.

## 3.1 Three-particle amplitudes

We want to discuss some Universal structure of the three point amplitudes for massless particles.

### 3.1.1 Complex three-particle kinematics and little-group scaling

For three massless external momenta,

$$p_1 + p_2 + p_3 = 0, \quad p_i^2 = 0. \tag{201}$$

Momentum conservation implies

$$0 = p_1^2 = (-p_2 - p_3)^2 = 2p_2 \cdot p_3, \tag{202}$$

and similarly for the other pairs. In spinor-helicity variables,

$$2p_i \cdot p_j = \langle ij\rangle[ji]. \tag{203}$$

For real momenta, $\langle ij\rangle^* = [ji]$, so the momentum conservation condition implies that:

$$\langle 12\rangle = \langle 23\rangle = \langle 31\rangle = 0, \quad [12] = [23] = [31] = 0. \tag{204}$$

Therefore if some Feymann rules tells us that the three-point function built from such spinors. Then it is zero.

> **Remark**
>
> An example is the 3 point gluon vertex in Yang-Mills theory. The three point function is zero for real momenta. However, this doesn't mean that all 3 point functions are zero. For example, the $\varphi^3$ theory it has a well defined $-ig$ three point amplitude.

For complex momenta the angle and square spinors are independent, and there are two possible branches. Assume, for example, that $\langle 23\rangle \neq 0$. Then

$$\langle 23\rangle[32] = 0 \quad \rightarrow \quad [32] = 0. \tag{205}$$

Moreover,

$$0 = \langle 2|\ (p_1 + p_2 + p_3)\ |1] = \langle 23\rangle[31], \tag{206}$$

so $[31] = 0$; similarly $[12] = 0$. Hence

$$[12] = [23] = [31] = 0, \tag{207}$$

and all square spinors $|i]$ are proportional. Conversely, if $[23] \neq 0$, then

$$\langle 12\rangle = \langle 23\rangle = \langle 31\rangle = 0, \tag{208}$$

and all angle spinors $|i\rangle$ are proportional. Thus if we promote the outgoing momentum to be non-zero, three-point amplitudes that used to be 0 due to the kinetic constraints may be non-zero and are either holomorphic or anti-holomorphic.

Under little-group transformations, remember that the spinors transform as:

$$|i\rangle \to t_i|i\rangle, \quad |i] \to t_i^{-1}\ |i], \quad t = e^{i\frac{\phi}{2}} \tag{209}$$

where $\phi$ is the angle rotated around the momentum direction.

For a three-point amplitude, we assume that it lives consistently on one of the two branches. And the external state of the amplitude have the helicity of $h_i$. This means that under little group transformation, each external point gain phase factor of:

$$e^{-ih_i\phi} \tag{210}$$

A generic assumption for the spinor dependence of the tree level three-point amplitude is:

$$\mathcal{A}_3\left(1^{h_1}, 2^{h_2}, 3^{h_3}\right) = \langle 12\rangle^{x_{12}}\langle 23\rangle^{x_{23}}\langle 31\rangle^{x_{31}}. \tag{211}$$

Both sides should have the same little group transformation rules, therefore:

$$-2h_1 = x_{12} + x_{31}, \quad -2h_2 = x_{12} + x_{23}, \quad -2h_3 = x_{23} + x_{31}, \tag{212}$$

and therefore

$$\mathcal{A}_3 = \langle 12\rangle^{h_3-h_1-h_2}\langle 23\rangle^{h_1-h_2-h_3}\langle 31\rangle^{h_2-h_3-h_1}. \tag{213}$$

Similarly, if the amplitude lives on the other branch, we have:

$$\mathcal{A}_3 = [12]^{h_1+h_2-h_3}[23]^{-h_1+h_2+h_3}[31]^{h_1-h_2+h_3}. \tag{214}$$

Thus Lorentz invariance together with little-group scaling fixes the spinor structure of every three-point amplitude, up to coupling constants. Indeed we have some examples for this universal behavior.

- Yukawa Theory: $\mathcal{L} = y\varphi\overline{f}f$.

$$\mathcal{A}_3\left(f_1^-, \overline{f}_2^-, \varphi_3\right) = y\langle 12\rangle, \tag{215}$$

- QED:

$$\mathcal{A}_3\left(f_1^-, \overline{f}_2^+, \gamma_3^-\right) = -\sqrt{2}e\frac{\langle 13\rangle^2}{\langle 12\rangle}, \tag{216}$$

$$\mathcal{A}_3\left(f_1^-, \overline{f}_2^+, \gamma_3^+\right) = \sqrt{2}e\frac{[23]^2}{[12]}. \tag{217}$$

### 3.1.2 Three-vector amplitudes

- $(-,-,+)$ Helicity

If we assume that the three point ingoing particles have $(--+)$ helicity. Then the only possible three-point amplitude is:

$$\mathcal{A}_3(1^-, 2^-, 3^+) = g\frac{\langle 12\rangle^3}{\langle 23\rangle\langle 31\rangle}, \tag{218}$$

The first expression is odd under $1 \leftrightarrow 2$. Identical photons therefore have no such self-interaction. For non-Abelian gauge bosons the antisymmetry is carried by the color factor,

$$\mathcal{A}_3(1_a^-, 2_b^-, 3_c^+) = gf^{abc}\frac{\langle 12\rangle^3}{\langle 23\rangle\langle 31\rangle}, \quad f^{abc} = -f^{bac}. \tag{219}$$

This makes this amplitude to possibly exist.

- $(-,+,+)$ Helicity

$$\mathcal{A}_3(1^-, 2^+, 3^+) = gf^{abc}\frac{[23]^3}{[12][31]}. \tag{220}$$

We take the antiholomorphic branch due to potential inconsistency with the holomorphic branch that violates locality.

- $(-,-,-)$ Helicity

The solution may take the following form,

$$\mathcal{A}_3(1^-,2^-,3^-)=\frac{1}{\Lambda^2}\langle 12\rangle\langle 23\rangle\langle 31\rangle, \tag{221}$$

However, one may notice that such interaction is not renormalizable.

One can find that for generic spin-1 bosons, the only possible three-point amplitudes are the ones listed above:

$$\mathcal{A}_3(1^-,2^+,3^+)=gf^{abc}\frac{[23]^3}{[12][31]}.\quad \mathcal{A}_3(1_a^-,2_b^-,3_c^+)=gf^{abc}\frac{\langle 12\rangle^3}{\langle 23\rangle\langle 31\rangle}, \tag{222}$$

One may find them exactly match the three-point amplitudes from Yang-Mills theory.

## 3.2 BCFW recursion

Here we focus on the on shell amplitudes of massless particles. An amazing fact is that higher-point tree amplitudes can be reconstructed from lower-point on-shell amplitudes.

### 3.2.1 General complex deformation and factorization

For a tree level amplitude of $n$ massless particles, we assume all ingoing. Thus the exernal momenta satisfy:

$$\sum_{i=1}^{n}p_i=0,\quad p_i^2=0. \tag{223}$$

Introduce a complex deformation of the external momenta,

$$\hat{p}_i(z)=p_i+zr_i, \tag{224}$$

with complex values vectors $r_i$ (certainly it can be 0) satisfying

$$r_i\cdot r_j=0\quad \text{for all } i,j,\quad p_i\cdot r_i=0,\quad \sum_i r_i=0. \tag{225}$$

These conditions preserve the on-shell constraints and total momentum conservation:

$$\hat{p}_i^2=0,\quad \sum_i\hat{p}_i=\sum_i p_i=0 \tag{226}$$

For a factorization channel $I$ (a subset of the external legs) that have at least 2 no more than n-2 external legs that are non-trivial ($P_I^2\neq 0$), define:

$$P_I=\sum_{i\in I}p_i,\quad R_I=\sum_{i\in I}r_i,\quad \hat{P}_I(z)=P_I+zR_I. \tag{227}$$

The shifted propagator goes on shell at

$$\hat{P}_I^2(z)=P_I^2+2zP_I\cdot R_I=-P_I^2\frac{z-z_I}{z_I},\quad z_I=-\frac{P_I^2}{2P_I\cdot R_I}. \tag{228}$$

Now we can do a shift to the amplitude. The shifted amplitude is defined as:

$$\hat{\mathcal{A}}_n(z)=\mathcal{A}_n(\hat{p}_1(z),...,\hat{p}_n(z)). \tag{229}$$

> **Remark**
>
> Here we only focus on tree level amplitudes. For loop it will be more complicated. It is also important to note that not only the momentum is shifted. The spinors built from momentum and polarization vectors built from momentum are both shifted.

The shifted tree amplitude $\hat{\mathcal{A}}_n(z)$ is a function of $z$. Apparantly, if we take $z = 0$, we recover the original amplitude:

$$\hat{\mathcal{A}}_n(0) = \mathcal{A}_n. \tag{230}$$

We now examine the analytic structure of $\hat{\mathcal{A}}_n(z)$ as a function of $z$. At tree level, $\hat{\mathcal{A}}_n(z)$ is a rational function of $z$: it has no branch cuts, logarithms, or square roots. Its singularities are therefore poles. For generic external momenta, these poles are simple.

To see this, consider the Feynman diagrams. A finite pole in $z$ can arise when a shifted internal propagator goes on shell. For a nontrivial subset $I$ of external legs, define

$$\hat{P}_I(z) = \sum_{i \in I} \hat{p}_i(z). \tag{231}$$

If $P_I \cdot R_I \neq 0$, the propagator is

$$\frac{1}{\hat{P}_I^2(z)} = -\frac{z_I}{P_I^2(z - z_I)}, \quad z_I = -\frac{P_I^2}{2P_I \cdot R_I}. \tag{232}$$

Thus it has a simple pole at $z = z_I$, which is away from the origin for generic external momenta. In a tree diagram, a given internal propagator occurs only once. Moreover, for generic momenta, different factorization channels have distinct values of $z_I$. Consequently, $\hat{\mathcal{A}}_n(z)$ has only simple finite poles.

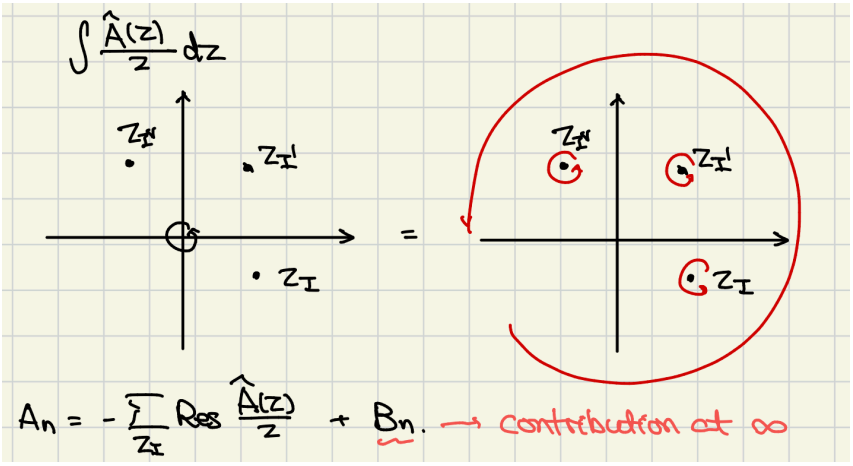



Figure 1: Illustration of the contour integral in the complex $z$ plane. The contour is deformed to encircle all finite poles, and a large circle at infinity.

Therefore, Cauchy's theorem gives

$$\mathcal{A}_n = -\sum_{z_I} \text{Res}_{z=z_I} \frac{\hat{\mathcal{A}}_n(z)}{z} + B_n, \tag{233}$$

where $B_n$ is the contribution from $z = \infty$. Near a physical pole, tree-level factorization implies

$$\hat{\mathcal{A}}_n(z) \to \hat{\mathcal{A}}_L(z_I) \frac{1}{\hat{P}_I^2(z)} \hat{\mathcal{A}}_R(z_I), \tag{234}$$

and

$$\frac{1}{\hat{P}_I^2(z)} = -\frac{z_I}{z - z_I}\frac{1}{P_I^2}. \tag{235}$$

If the deformation is chosen so that $B_n = 0$, the BCFW recursion relation is

$$\mathcal{A}_n = \sum_{I,h} \hat{\mathcal{A}}_L(z_I)\frac{1}{P_I^2}\hat{\mathcal{A}}_R(z_I). \tag{236}$$

The sum runs over factorization channels, for each channel, all legs in $I$ should be either in the left part or in the right part. However, should not double count!! One should also remember to sum over the helicities and other DoFs of the inner leg. The subamplitudes on both sides are evaluated at $z = z_I$. We may find these subamplitudes are on shell amplitudes with fewer external legs and complex momenta.

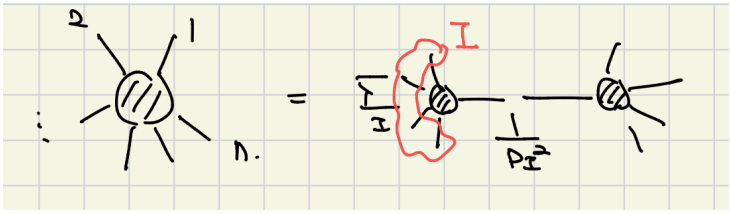



Figure 2: Illustration

### 3.2.2 BCFW Recursion Relation

Notice that the shift $r_i$ may be taken to be 0. For a BCFW recursion relation, we choose only two external legs to shift. And the shift shall take the following explicit form:

$$|\hat{i}] = |i] + z\,|j], \quad |\hat{j}\rangle = |j\rangle - z|i\rangle, \quad |\hat{i}\rangle = |i\rangle, \quad |\hat{j}] = |j]. \tag{237}$$

Equivalently,

$$\hat{p}_i = p_i + z|i\rangle[j|, \quad \hat{p}_j = p_j - z|i\rangle[j|. \tag{238}$$

For a negative-helicity gauge boson on leg $i$ and a positive-helicity gauge boson on leg $j$, both polarization vectors scale as $z^{-1}$. The part of a tree diagram connecting the two shifted legs grows at worst as $z$, hence

$$\hat{\mathcal{A}}_n(z) = O(z^{-1}) \quad \text{as} \quad z \to \infty, \tag{239}$$

so $B_n = 0$. Therefore one can see that having $B_n = 0$ only need a certain choice of the shift with the right helicity configuration of the external legs.

### 3.2.3 Example: Compton scattering

We consider the QED amplitude of compton scattering while taking the electrons and positrons to be massless.

- BCFW Shift of two Photons

Consider the QED amplitude

$$\mathcal{A}_4\left(f_1^-, \overline{f}_2^+, \gamma_3^-, \gamma_4^+\right) \tag{240}$$

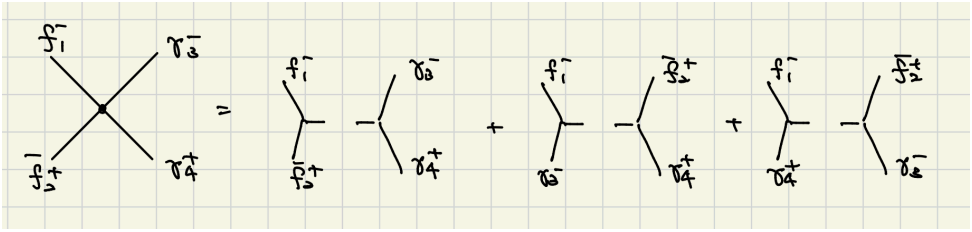
and use the $[3, 4\rangle$ shift

$$|\hat{3}] = |3] + z|4], \quad \left|\hat{4}\right\rangle = |4\rangle - z|3\rangle. \tag{241}$$

For only 4 external legs, we only have 3 Factorization channels, they are explicitly:

$$\hat{P} = p_1 + \hat{p}_3, \quad \hat{P} = p_2 + \hat{p}_4, \quad \hat{P} = p_1 + p_2. \tag{242}$$

Drawn in a diagram it is:



two factorization channels that may give a pole, the third channel has shifted propagator independent of $z$ therefore won't contribute to a pole. The channel that constributed have the momentum as following.

$$\hat{P} = p_2 + \hat{p}_4, \quad \hat{P} = p_1 + \hat{p}_4. \tag{243}$$

We first calculate the first channel. The first factorization channel has internal momentum $\hat{P} = p_2 + \hat{p}_4$. Its pole is fixed by

$$\hat{P}^2 = \langle 2\hat{4}\rangle [42] = 0, \quad z_{24} = \frac{\langle 24\rangle}{\langle 23\rangle}. \tag{244}$$

At the pole,

$$\langle 1\hat{4}\rangle = \langle 14\rangle - z_{24}\langle 13\rangle = \frac{\langle 14\rangle\langle 23\rangle - \langle 13\rangle\langle 24\rangle}{\langle 23\rangle} = -\frac{\langle 12\rangle\langle 34\rangle}{\langle 23\rangle}. \tag{245}$$

Then we calculate the three point amplitudes on both sides:

$$\mathcal{A}_{24} = A_L(z_{24})\frac{1}{P_{24}^2}A_R(z_{24}) \tag{246}$$

Using the result of the complex three-point amplitude of QED, we immediately have:

The two BCFW residues are

$$\mathcal{A}_{24} = 2e^2\frac{\langle 13\rangle^2\langle 23\rangle}{\langle 24\rangle\langle 12\rangle\langle 34\rangle}, \tag{247}$$

$$\mathcal{A}_{14} = -2e^2\frac{\langle 13\rangle^3}{\langle 14\rangle\langle 12\rangle\langle 34\rangle}. \tag{248}$$

Adding them,

$$\mathcal{A}_4 = -2e^2\frac{\langle 13\rangle^2}{\langle 12\rangle\langle 34\rangle}\left(\frac{\langle 13\rangle}{\langle 14\rangle} - \frac{\langle 23\rangle}{\langle 24\rangle}\right). \tag{249}$$

Using the Schouten identity,

$$\frac{\langle 13\rangle}{\langle 14\rangle} - \frac{\langle 23\rangle}{\langle 24\rangle} = \frac{\langle 12\rangle\langle 34\rangle}{\langle 14\rangle\langle 24\rangle}, \tag{250}$$

one obtains

$$\mathcal{A}_4\left(f_1^-, \overline{f}_2^+, \gamma_3^-, \gamma_4^+\right) = -2e^2\frac{\langle 13\rangle^2}{\langle 14\rangle\langle 24\rangle}. \tag{251}$$

- BCFW Shift of a Photon and a Positron

A different valid shift can produce the same result from a single on-shell diagram. Equality of different BCFW representations is a useful consistency condition.

#### 3.2.4 Adding color: the Lie algebra from on-shell consistency

Let the fermion-fermion-vector three-point amplitude be

$$\mathcal{A}_3\big(f_i^-, \overline{f}_j^+, g_A^-\big) = T^A{}_{ij}\frac{\langle 13\rangle^2}{\langle 12\rangle}. \tag{252}$$

Computing the colored Compton amplitude from the two fermion-exchange channels gives

$$\mathcal{A}_4 = \frac{\langle 13\rangle^2}{\langle 12\rangle\langle 34\rangle}\left(T^A{}_{ik}T^B{}_{kj}\frac{\langle 23\rangle}{\langle 24\rangle} - T^B{}_{ik}T^A{}_{kj}\frac{\langle 13\rangle}{\langle 14\rangle}\right). \tag{253}$$

With another shift, one also exposes the gauge-boson exchange channel:

$$\mathcal{A}_4 = \frac{\langle 13\rangle^2}{\langle 24\rangle\langle 41\rangle}T^A{}_{ik}T^B{}_{kj} - f_{ABC}T^C{}_{ij}\frac{\langle 13\rangle^3}{\langle 12\rangle\langle 34\rangle\langle 41\rangle}. \tag{254}$$

Requiring both constructions to agree forces the couplings to close under a Lie algebra. With Hermitian generators,

$$T^A{}_{ik}T^B{}_{kj} - T^B{}_{ik}T^A{}_{kj} = if^{ABC}T^C{}_{ij}, \tag{255}$$

or equivalently

$$\big[T^A, T^B\big] = if^{ABC}T^C. \tag{256}$$

If the factor $i$ is absorbed into the definition of the cubic coupling, the same relation is written without the explicit $i$. Thus locality, factorization, and shift independence already imply the Lie-algebra structure of the gauge couplings.

### 3.3 Vanishing sectors, MHV amplitudes, and Parke-Taylor

#### 3.3.1 Helicity selection and color ordering

At tree level, pure Yang-Mills amplitudes with all equal helicities vanish. The same is true when only one gluon has the opposite helicity:

$$A_n(1^-, 2^-, ..., n^-) = 0, \quad A_n(1^+, 2^-, ..., n^-) = 0. \tag{257}$$

Parity gives the corresponding all-plus and one-minus statements. These results follow inductively from BCFW recursion because every possible factorization contains a vanishing lower-point amplitude.

The first nonzero generic tree amplitudes have two negative-helicity gluons and are called MHV amplitudes:

$$A_{n(1^-,2^-,3^+,...,n^+)}. \tag{258}$$

The full Yang-Mills amplitude can be decomposed into color-ordered partial amplitudes,

$$\mathcal{A}_n(1^{a_1}, ..., n^{a_n}) = g^{n-2}\sum_{\sigma\in\frac{S_n}{Z_n}} A_n(\sigma(1), ..., \sigma(n))\ \mathrm{tr}(T^{a_{\sigma(1)}}...T^{a_{\sigma(n)}}). \tag{259}$$

The partial amplitude $A_n$ is associated with a fixed cyclic ordering. Its physical poles occur only when adjacent ordered momenta become collinear or when sums of cyclically adjacent momenta go on shell.

### 3.3.2 Four-point MHV amplitude and the n-point formula

For the ordering $(1,2,3,4)$, BCFW recursion gives, in the overall sign convention of the handwritten notes,

$$A_4(1^-,2^-,3^+,4^+) = -g^2 \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\langle 34\rangle\langle 41\rangle}. \tag{260}$$

The denominator displays poles only for adjacent particles in the color ordering, while the numerator is fixed by little-group scaling. Suppressing the overall coupling and convention-dependent phase suggests the $n$-point ansatz

$$A_{n(1^-,2^-,3^+,\ldots,n^+)} = \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\ldots\langle n-1,n\rangle\langle n1\rangle}. \tag{261}$$

This formula follows by induction. Apply the $[1,n\rangle$ shift,

$$|\hat{1}] = |1] + z\,|n], \quad |\hat{n}\rangle = |n\rangle - z|1\rangle. \tag{262}$$

For the chosen ordering and helicities, only the channel

$$\hat{P} = p_{n-1} + \hat{p}_n \tag{263}$$

contributes. The on-shell condition gives

$$\langle n-1,\hat{n}\rangle = 0, \quad z_* = \frac{\langle n-1,n\rangle}{\langle n-1,1\rangle}. \tag{264}$$

At this pole, the Schouten identity gives

$$\langle n-2,\hat{n}\rangle = \frac{\langle n-2,n\rangle\langle n-1,1\rangle - \langle n-2,1\rangle\langle n-1,n\rangle}{\langle n-1,1\rangle} = \frac{\langle n-2,n-1\rangle\langle 1n\rangle}{\langle n-1,1\rangle}. \tag{265}$$

Substituting the induction hypothesis for the left subamplitude and the appropriate three-point amplitude on the right yields

$$A_{n(1^-,2^-,3^+,\ldots,n^+)} = \frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\ldots\langle n-1,n\rangle\langle n1\rangle}. \tag{266}$$

More generally, if the two negative-helicity legs are $r$ and $s$,

$$A_{n(1^+,\ldots,r^-,\ldots,s^-,\ldots,n^+)} = \frac{\langle rs\rangle^4}{\prod_{a=1}^{n}\langle a,a+1\rangle}, \quad n+1 \equiv 1. \tag{267}$$

This is the Parke-Taylor formula for color-ordered tree-level gluon amplitudes.

Amplitudes with three negative helicities are called NMHV; more generally, amplitudes with $k+2$ negative-helicity gluons are $\mathrm{N}^k$ MHV. BCFW recursion expresses each sector in terms of lower-point amplitudes. In QCD this classification is fundamentally a tree-level organization; loop amplitudes have a richer structure.

# 4 Path Integral Formulation

## 4.1 From canonical quantisation to path integrals

Our goal is to describe the dynamics of a quantum system. In canonical quantisation, we start from a classical Lagrangian, identify conjugate momenta, and impose commutation relations. The path integral is another route from the classical theory to the quantum theory.

The two routes are expected to agree where both are defined. Derivative interactions can make canonical momenta difficult to identify. Anomalies, solitons, instantons, and other nonperturbative phenomena are also naturally discussed with path integrals. We first derive the path integral in a quantum-mechanical example and then use the result for field theory.

Schematically, the two constructions start from the same classical Lagrangian:

$$\mathcal{L}_{\text{classical}} \to \text{QFT (canonical quantisation)}, \quad \mathcal{L}_{\text{classical}} \to \text{QFT (path integral)}. \tag{268}$$

### 4.1.1 Quantum mechanics

Consider coordinates $q_i$ for $i = 1, ..., N$, with

$$L = \frac{m}{2}\sum_{i=1}^{N} \dot{q}_i^2 - V(q_1, ..., q_N), \quad p_i = \frac{\partial L}{\partial \dot{q}_i} = m\dot{q}_i, \quad H = \sum_{i=1}^{N} \frac{p_i^2}{2m} + V(q). \tag{269}$$

Promote $q_i, p_i$ to operators satisfying

$$\left[\hat{q}_i, \hat{p}_j\right] = i\hbar\delta_{ij}, \quad \left[\hat{q}_i, \hat{q}_j\right] = \left[\hat{p}_i, \hat{p}_j\right] = 0. \tag{270}$$

In the Schrödinger picture, the transition amplitude between configurations at $-T$ and $T$ is

$$K(q^{\text{out}}, T; q^{\text{in}}, -T) = \left\langle q^{\text{out}} \right| e^{-2iT\frac{\hat{H}}{\hbar}} \left| q^{\text{in}} \right\rangle. \tag{271}$$

To see the role of intermediate configurations, write $2T = I\varepsilon$ and insert position-space resolutions of the identity between short evolution operators. A discretised path is specified by

$$q_i^k = q_{i(-T+(k-1)\varepsilon)}, \quad q_i^1 = q_i^{\text{in}}, \quad q_i^{I+1} = q_i^{\text{out}}. \tag{272}$$

For the kinetic term, the short-time kernel is Gaussian. Multiplying the kernels and then taking $I \to \infty$ gives the action weighted by $e^{i\frac{S}{\hbar}}$.

More explicitly, to leading order in $\varepsilon$,

$$\left\langle q^{k+1} | e^{-i\varepsilon\frac{\hat{H}}{\hbar}} | q^k \right\rangle \approx A^{-N} \exp \frac{i\varepsilon}{\hbar} \frac{m}{2} \sum_{i=1}^{N} \frac{q_i^{k+1} - q_i^k}{\varepsilon}^2 - V(q^k), \tag{273}$$

and the sum of the exponents tends to the action,

$$S[q] = \lim_{I\to\infty} \sum_{k=1}^{I} \varepsilon \frac{m}{2} \sum_{i=1}^{N} \frac{q_i^{k+1} - q_i^k}{\varepsilon}^2 - V(q^k) = \int_{-T}^{T} \mathrm{d}t L(q, \dot{q}). \tag{274}$$

Slicing the time interval and inserting complete sets of position states leads to

$$K(q^{\text{out}}, T; q^{\text{in}}, -T) = \int \mathcal{D}q \exp \frac{i}{\hbar} \int_{-T}^{T} \mathrm{d}t L(q, \dot{q}) = \int \mathcal{D}q e^{i\frac{S[q]}{\hbar}}. \tag{275}$$

Here every path has endpoints $q(-T) = q^{\text{in}}$ and $q(T) = q^{\text{out}}$. The action depends on the entire path. With time step $\varepsilon = 2\frac{T}{I}$, an illustrative definition of the measure is

$$\mathcal{D}q = \lim_{I\to\infty} \prod_{i=1}^{N} \prod_{k=2}^{I} \frac{\mathrm{d}q_i^k}{A}, \quad A = \sqrt{\frac{2\pi i\hbar\varepsilon}{m}}. \tag{276}$$

The first and last time slices are fixed by the boundary conditions. The overall normalisation of the measure will cancel in normalised correlators.

The notation $\mathcal{D}q$ means that one integrates over the interior value of every component at every time slice. The path integral is a limit of ordinary integrals; the path itself, rather than only one value of $q$, is the integration variable.

### 4.1.2 Euclidean path integral

Real-time integrals contain rapidly oscillating phases, of the type $\int \mathrm{d}x e^{ix^2}$. They can be defined by analytic continuation from convergent Gaussian integrals such as $\int \mathrm{d}x e^{-x^2}$. The corresponding field-theory construction starts with the Euclidean path integral.

Introduce Euclidean time $t = -i\tau$. Evolution by Euclidean time $2T_E$ is $e^{-2T_E \frac{H}{\hbar}}$, and

$$K_E(q^{\mathrm{out}}, T_E; q^{\mathrm{in}}, -T_E) = \left\langle q^{\mathrm{out}} | e^{-2T_E \frac{\hat{H}}{\hbar}} | q^{\mathrm{in}} \right\rangle = \int \mathcal{D}q e^{-\frac{S_{E[q]}}{\hbar}}, \tag{277}$$

where

$$S_{E[q]} = \int_{-T_E}^{T_E} \mathrm{d}\tau \frac{m}{2} \frac{\mathrm{d}q}{\mathrm{d}\tau}^2 + V(q). \tag{278}$$

The paths satisfy $q(-T_E) = q^{\mathrm{in}}$ and $q(T_E) = q^{\mathrm{out}}$. For this example, substituting $t = -i\tau$ in the classical action $S = \int \mathrm{d}t L$ gives

$$t \to -i\tau, \quad \frac{\mathrm{d}}{\mathrm{d}t} \to i\frac{\mathrm{d}}{\mathrm{d}\tau}, \tag{279}$$

$$S = \int \mathrm{d}t \frac{m}{2} \frac{\mathrm{d}q}{\mathrm{d}t}^2 - V(q) \to i \int \mathrm{d}\tau \frac{m}{2} \frac{\mathrm{d}q}{\mathrm{d}\tau}^2 + V(q) = iS_E. \tag{280}$$

Hence

$$S_E = -iS|_{t=-i\tau}. \tag{281}$$

For a general theory we use the same Wick rotation to define $S_E$. Real-time amplitudes are recovered by analytic continuation, with the appropriate ordering prescription.

### 4.1.3 Correlation functions and vacuum projection

The Heisenberg operator is

$$\hat{q}_{H,i}(t) = e^{i\hat{H}\frac{t}{\hbar}} \hat{q}_i e^{-i\hat{H}\frac{t}{\hbar}}. \tag{282}$$

For the lowest-energy state $|0\rangle$, define

$$G_{i_1 \ldots i_n}(t_1, \ldots, t_n) = \left\langle 0 | \mathcal{T}\left[\hat{q}_{H,i_1}(t_1) \ldots \hat{q}_{H,i_n}(t_n)\right] | 0 \right\rangle. \tag{283}$$

For ordered Euclidean times $\tau_1 > \tau_2 > \ldots > \tau_n$, the continuation of the operator product contains positive Euclidean evolution intervals:

$$\left\langle 0 | \hat{q}_{i_1}(\tau_1) \ldots \hat{q}_{i_n}(\tau_n) | 0 \right\rangle = \left\langle 0 | e^{\tau_1 \frac{H}{\hbar}} \hat{q}_{i_1} e^{-(\tau_1 - \tau_2)\frac{H}{\hbar}} \hat{q}_{i_2} \ldots \hat{q}_{i_n} e^{-\tau_n \frac{H}{\hbar}} | 0 \right\rangle. \tag{284}$$

These factors suppress higher-energy intermediate states when successive time separations are positive. Euclidean time ordering is defined explicitly by

$$\mathcal{T}[A(\tau_1)B(\tau_2)] = \begin{cases} A(\tau_1)B(\tau_2) \text{ if } \tau_1 > \tau_2 \\ B(\tau_2)A(\tau_1) \text{ if } \tau_2 > \tau_1 \end{cases}. \tag{285}$$

Euclidean time ordering places the operator with the larger $\tau$ to the left. The Euclidean correlator is

$$G_{E;i_1\ldots i_n}(\tau_1, ..., \tau_n) = \frac{\int \mathcal{D}q q_{i_1}(\tau_1)...q_{i_n}(\tau_n) e^{-\frac{S_{E[q]}}{\hbar}}}{\int \mathcal{D}q e^{-\frac{S_{E[q]}}{\hbar}}}. \tag{286}$$

The ratio removes the overall normalisation of the measure. Taking the initial and final Euclidean times to infinity projects onto the vacuum, provided the boundary states have nonzero overlap with it. Indeed, if $\langle 0|q^{\text{in}}\rangle \neq 0$ and $\langle q^{\text{out}}|0\rangle \neq 0$, then the lowest-energy contribution dominates the factors $e^{-T_E \frac{H}{\hbar}}$. The boundary values can therefore be chosen conveniently in the infinite-time limit.

The correlator in real time follows by analytic continuation of the ordered Euclidean correlator. The ordering of the operators is retained during that continuation.

## 4.2 A scalar field theory

Put a real scalar field on a cubic spatial lattice with sites $\boldsymbol{x}_n = a\boldsymbol{n}$, where $n_i = -N, ..., N$. At fixed time, the values

$$\varphi_{n(t)} = \varphi(x_n, t) \tag{287}$$

play the role of the coordinates $q_{i(t)}$ above. For a theory without derivative interactions, the Lagrangian is

$$L = \int \mathrm{d}^3x \frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi - V(\varphi) = \int \mathrm{d}^3x \frac{1}{2}(\partial_0\varphi)^2 - \frac{1}{2}\sum_{i=1}^{3}(\partial_i\varphi)^2 - V(\varphi). \tag{288}$$

The lattice Lagrangian is, with the spatial differences displayed explicitly,

$$L_a = a^3 \sum_n \frac{1}{2}\dot{\varphi}_n^2 - \frac{1}{2}\sum_{i=1}^{3}\frac{\varphi_{n+\hat{e}_i} - \varphi_n}{a}^2 - V(\varphi_n). \tag{289}$$

There are $(2N+1)^3$ coordinates $\varphi_{n(t)}$. The spatial differences and $V(\varphi)$ act as a potential for these coordinates, so the quantum-mechanical path integral applies directly. Its Euclidean action before the continuum limit is

$$S_{E,a} = \int \mathrm{d}\tau a^3 \sum_n \frac{1}{2}(\partial_\tau\varphi_n)^2 + \frac{1}{2}\sum_{i=1}^{3}\frac{\varphi_{n+\hat{e}_i} - \varphi_n}{a}^2 + V(\varphi_n). \tag{290}$$

Taking $a \to 0$ and $N \to \infty$ while keeping the spatial size under control gives

$$S_{E[\varphi]} = \int \mathrm{d}\tau \mathrm{d}^3x \frac{1}{2}(\partial_\tau\varphi)^2 + \frac{1}{2}\sum_{i=1}^{3}(\partial_i\varphi)^2 + V(\varphi). \tag{291}$$

With Euclidean coordinates $x_E = (\tau, \boldsymbol{x})$ and metric $\delta_{\mu\nu}$, this becomes

$$S_{E[\varphi]} = \int \mathrm{d}^4x_E \frac{1}{2}\partial^E_\mu\varphi\partial^E_\mu\varphi + V(\varphi). \tag{292}$$

For the Minkowski metric $(+,-,-,-)$, Wick rotation sends $x^0 = t \to -i\tau$ and $\partial_0 \to i\partial_\tau$. The Euclidean kinetic term is positive. The formal field measure is a limit of ordinary integrals over every lattice site and Euclidean time slice:

$$\mathcal{D}\varphi \propto \lim_{a\to 0}\lim_{\varepsilon\to 0}\prod_{n,k}\mathrm{d}\varphi_n^k. \tag{293}$$

For later reference, the transformation of spacetime components is

$$x^0 \to -ix_E^0, \quad x^i \to x_E^i, \quad x_0 \to -ix_{E,0}, \quad x_i \to -x_{E,i}, \tag{294}$$

$$\partial_0 \to i\partial_0^E, \quad \partial_i \to \partial_i^E. \tag{295}$$

For example, after integration by parts the Minkowski kinetic term $\partial_\mu\varphi\partial^\mu\varphi$ is $-\varphi\Box\varphi$; its continuation gives the positive Euclidean quadratic form $\varphi(-\Box_E)\varphi$.

The calculation of vacuum correlators follows three steps: choose the classical Lagrangian, derive $S_E$ by Wick rotation, and evaluate normalised Euclidean path integrals before analytic continuation.

The continuum measure is formal, but lattice regularisation makes it a finite-dimensional integral before the limits are taken. Lattice QFT can address questions that perturbation theory cannot. A lattice theory can also be defined at finite cutoff even when its continuum high-energy limit does not exist.

For example, a higher-derivative interaction of the schematic form

$$\mathcal{L} = \frac{1}{2}\partial_\mu\varphi\partial^\mu\varphi + \varphi^2\big(\partial_\mu\varphi\partial^\mu\varphi\big)^4\Box\varphi \tag{296}$$

can be inserted into the path integral without first constructing an interacting canonical momentum. At fixed lattice spacing, the discretised Euclidean path integral is meaningful for a suitable stable action; taking a continuum limit is a separate question.

### 4.2.1 Vector fields

For a Lorentz vector, the Euclidean components must also be continued. With one common index convention,

$$A_0 \to iA_{E,0}, \quad A_i \to A_{E,i}. \tag{297}$$

Equivalently, for contravariant components one uses $A^0 \to iA_E^0$ and $A^i \to -A_E^i$ under this convention. This must be combined with the continuation of the derivatives when constructing $F_{E,\mu\nu}$.

Using the corresponding continuation of the coordinates and derivatives, the Minkowski Maxwell action

$$S = -\frac{1}{4}\int \mathrm{d}^4x F_{\mu\nu}F^{\mu\nu} \tag{298}$$

becomes

$$F_{0i} \to iF_{E,0i}, \quad F_{ij} \to F_{E,ij}, \quad F_{\mu\nu}F^{\mu\nu} = -2\sum_i F_{0i}^2 + \sum_{i,j} F_{ij}^2, \tag{299}$$

$$-iS|_{t=-i\tau} = \frac{1}{4}\int \mathrm{d}^4x_E 2\sum_i F_{E,0i}^2 + \sum_{i,j} F_{E,ij}^2. \tag{300}$$

Thus

$$S_E = \frac{1}{4}\int \mathrm{d}^4x_E F_{E,\mu\nu}F_{E,\mu\nu}. \tag{301}$$

In particular, the Minkowski contraction $F_{\mu\nu}F^{\mu\nu}$ has different signs for electric and magnetic components, whereas the Euclidean contraction is a sum of squares.

## 4.3 Free theories and generating functionals

Split the Euclidean action into a part we can solve and an interaction:

$$S_E = S_0 + S_{\text{int}}. \tag{302}$$

The solvable part is quadratic,

$$S_0[\varphi] = \frac{1}{2}\int \mathrm{d}^4x\mathrm{d}^4y\varphi(x)K_{xy}\varphi(y). \tag{303}$$

Here $K_{xy}$ may contain derivatives acting on the field. Calling $S_0$ quadratic means that every term contains exactly two fields; it does not imply that every quadratic term of the full action must be placed in $S_0$. We choose the split so that the Gaussian integral is well defined and the remainder can be expanded perturbatively. For local theories,

$$K_{xy} = K_x\delta^4(x-y). \tag{304}$$

For a local massive scalar,

$$S_0 = \frac{1}{2}\int \mathrm{d}^4x\partial_\mu\varphi\partial_\mu\varphi + m^2\varphi^2 = \frac{1}{2}\int \mathrm{d}^4x\varphi(x)(m^2-\Box_x)\varphi(x), \tag{305}$$

so $K_x = m^2 - \Box_x$ and $K_{xy} = K_x\delta^4(x-y)$.

### 4.3.1 Gaussian integral

Introduce a source $J(x)$:

$$Z_0[J] = \int \mathcal{D}\varphi \exp -S_0[\varphi] + \int \mathrm{d}^4x J(x)\varphi(x). \tag{306}$$

Functional derivatives generate correlators:

$$\langle\varphi(x_1)...\varphi(x_n)\rangle_0 = \frac{1}{Z_0[0]}\frac{\delta^n Z_0[J]}{\delta J(x_1)...\delta J(x_n)}_{J=0}. \tag{307}$$

The finite-dimensional Gaussian identity behind this result is

$$\int \mathrm{d}^N q e^{-\frac{1}{2}q^TAq+b^Tq} = Z_0[0]e^{\frac{1}{2}b^TA^{-1}b}, \tag{308}$$

up to a measure-dependent normalisation. It requires a positive quadratic form. Let $D = K^{-1}$, meaning

$$-\frac{1}{2}q^TAq + b^Tq = -\frac{1}{2}(q-A^{-1}b)^TA(q-A^{-1}b) + \frac{1}{2}b^TA^{-1}b. \tag{309}$$

Shifting the integration variable leaves a source-independent Gaussian integral. The same calculation in field space uses the inverse kernel $D = K^{-1}$, meaning

$$\int \mathrm{d}^4z K_{xz}D_{zy} = \delta^4(x-y). \tag{310}$$

The positivity of $K$ matters: a negative eigenvalue gives an exponentially unbounded direction of integration. If this happens, the chosen quadratic action does not provide a stable Gaussian starting point; additional terms in $S_E$ may be needed.

Completing the square gives

$$Z_0[J] = Z_0[0] \exp \frac{1}{2} \int \mathrm{d}^4x \mathrm{d}^4y J(x) D_{xy} J(y). \tag{311}$$

In particular, the Euclidean two-point function is

$$\langle \varphi(x)\varphi(y) \rangle_0 = \frac{1}{Z_0[0]} \left( \frac{\delta^2 Z_0[J]}{\delta J(x) \delta J(y)} \right)_{J=0} = D_{xy}. \tag{312}$$

### 4.3.2 Momentum-space propagator

Use the Fourier convention

$$f(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^4} e^{-ip\cdot x} \tilde{f}(p), \quad \tilde{f}(p) = \int \mathrm{d}^4x e^{ip\cdot x} f(x). \tag{313}$$

For a translation-invariant local operator $K_x$,

$$K_x e^{-ip\cdot(x-y)} = \tilde{K}(p) e^{-ip\cdot(x-y)}. \tag{314}$$

The delta function has the Fourier representation

$$\delta^4(x-y) = \int \frac{\mathrm{d}^4p}{(2\pi)^4} e^{-ip\cdot(x-y)}. \tag{315}$$

Substituting the Fourier ansatz for $D$ into $K_x D_{xy} = \delta^4(x-y)$ gives

$$\int \frac{\mathrm{d}^4p}{(2\pi)^4} e^{-ip\cdot(x-y)} \tilde{K}(p) \tilde{D}(p) = \int \frac{\mathrm{d}^4p}{(2\pi)^4} e^{-ip\cdot(x-y)}. \tag{316}$$

Thus

$$D_{xy} = \int \frac{\mathrm{d}^4p}{(2\pi)^4} e^{-ip\cdot(x-y)} \tilde{D}(p), \quad \tilde{D}(p) = \frac{1}{\tilde{K}(p)}. \tag{317}$$

For the massive scalar,

$$\tilde{K}(p) = p_E^2 + m^2, \quad \Delta_F^E(x-y) = \int \frac{\mathrm{d}^4p_E}{(2\pi)^4} \frac{e^{-ip_E\cdot(x_E-y_E)}}{p_E^2+m^2}. \tag{318}$$

The free generating functional determines every correlator. For example, differentiating it four times and setting $J=0$ gives

$$\frac{\delta Z_0}{\delta J(x_1)} = (D\cdot J)(x_1) Z_0[J], \tag{319}$$

$$\frac{\delta^2 Z_0}{\delta J(x_1)\delta J(x_2)} = D_{12} + (D\cdot J)(x_1)(D\cdot J)(x_2) Z_0[J]. \tag{320}$$

Two further derivatives produce three terms without sources at $J=0$:

$$\frac{\delta^3 Z_0}{\delta J(x_1)\delta J(x_2)\delta J(x_3)} = D_{12}(D\cdot J)(x_3) + D_{13}(D\cdot J)(x_2) + D_{23}(D\cdot J)(x_1) + \prod_{i=1}^{3}(D\cdot J)(x_i) Z_0[J]. \tag{321}$$

The fourth derivative at $J=0$ therefore gives

$$\langle \varphi(x_1)\varphi(x_2)\varphi(x_3)\varphi(x_4) \rangle_0 = D_{12}D_{34} + D_{13}D_{24} + D_{14}D_{23}, \quad D_{ij} = D_{x_i x_j}. \tag{322}$$

These are the Wick contractions of the canonical approach.

In general, odd correlators vanish for a centred Gaussian, while even correlators sum over all pairings of their insertion points.

### 4.3.3 Several bosonic fields

Let $\varphi_{M(x)}$ denote all components of the bosonic fields. For

$$S_0 = \frac{1}{2}\int \mathrm{d}^4x\mathrm{d}^4y\varphi_{M(x)}K_{Mx,Ny}\varphi_{N(y)}, \tag{323}$$

the source functional is

$$Z_0[J] = Z_0[0]\exp\frac{1}{2}\int \mathrm{d}^4x\mathrm{d}^4yJ_{M(x)}D_{Mx,Ny}J_{N(y)}, \tag{324}$$

where repeated component labels are summed and

$$\int \mathrm{d}^4zK_{Mx,Rz}D_{Rz,Ny} = \delta_{MN}\delta^4(x-y). \tag{325}$$

For local $K$, Fourier transformation reduces this to matrix inversion:

$$K_{Mx,Ny} = \delta^4(x-y)(K_x)_{MN}, \quad (K_x)_{MN}e^{-ip\cdot(x-y)} = \tilde{K}_{MN}(p)e^{-ip\cdot(x-y)}. \tag{326}$$

Writing

$$D_{Mx,Ny} = \int \frac{\mathrm{d}^4p}{(2\pi)^4}e^{-ip\cdot(x-y)}\tilde{D}_{MN}(p), \tag{327}$$

the inverse-kernel equation becomes

$$\tilde{K}_{MR}(p)\tilde{D}_{RN}(p) = \delta_{MN}, \quad \tilde{D}(p) = \tilde{K}(p)^{-1}. \tag{328}$$

## 4.4 Analytic continuation to Minkowski space

The free theory above was solved in Euclidean signature. To recover real-time correlators, we continue the time coordinate and choose a pole prescription. Consider a generic scalar kinetic polynomial $\tilde{K}(p_E^0, \boldsymbol{p})$:

$$\Delta_F^E(x_E) = \int \frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}\int_{\mathbb{R}}\frac{\mathrm{d}z}{2\pi}\frac{e^{-izx_E^0}}{\tilde{K}(z,\boldsymbol{p})}. \tag{329}$$

For a massive scalar,

$$\frac{1}{\tilde{K}(z,\boldsymbol{p})} = \frac{1}{z^2+E_p^2}, \quad E_p = \sqrt{\boldsymbol{p}^2+m^2}, \tag{330}$$

with poles at $z_+ = iE_p$ and $z_- = -iE_p$. More generally, when the kinetic polynomial depends on $p_E^2$, its zeros occur in pairs. Assuming no pole lies on the real Euclidean energy axis, the Euclidean energy integral is well defined.

Write

$$I_{E(x_E^0)} = \int_{\mathbb{R}}\frac{\mathrm{d}z}{2\pi}\frac{e^{-izx_E^0}}{\tilde{K}(z,\boldsymbol{p})}. \tag{331}$$

If $x_E^0 < 0$, the exponential decays when the contour is closed in the upper half-plane. For the two-pole example, only $z_+$ is enclosed, so the residue theorem gives

$$I_{E(x_E^0)} = i\mathrm{Res}_{z=z_+}\frac{e^{-izx_E^0}}{\tilde{K}(z,\boldsymbol{p})} = ie^{-ix_E^0 z_+}\mathrm{Res}_{z=z_+}\frac{1}{\tilde{K}(z,\boldsymbol{p})}. \tag{332}$$

Since

$$\mathrm{Res}_{z=iE_p}\frac{1}{z^2+E_p^2} = \frac{1}{2iE_p}, \tag{333}$$

we obtain

$$\int_{\mathbb{R}}\frac{\mathrm{d}z}{2\pi}\frac{e^{-izx_E^0}}{z^2+E_p^2} = \frac{e^{E_p x_E^0}}{2E_p} = \frac{e^{-E_p|x_E^0|}}{2E_p} \quad (x_E^0<0). \tag{334}$$

For $x_E^0>0$, close in the lower half-plane instead; together the two cases give $\frac{e^{-E_p|x_E^0|}}{2E_p}$. This exponential decay at large Euclidean time is the vacuum-projection behaviour discussed earlier.

For $x^0<0$, the analytic continuation $x_E^0 = ix^0$ gives

$$I_{M(x^0)} = I_{E(ix^0)} = \frac{e^{iE_p x^0}}{2E_p}, \tag{335}$$

and therefore

$$\Delta_{F(x)} = \int\frac{\mathrm{d}^3\boldsymbol{p}}{(2\pi)^3}e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}\frac{e^{iE_p x^0}}{2E_p} \quad (x^0<0). \tag{336}$$

The $x^0>0$ continuation yields the other time ordering. One can package both answers into a four-momentum integral by deforming the Euclidean energy contour so that it passes the poles in the Feynman way.

For a generic kinetic polynomial, define $\tilde{K}'(z') = \tilde{K}(-iz')$ and put $z' = iz$. Zeros $z_\pm$ of $\tilde{K}$ map to real-energy zeros $z'_\pm = iz_\pm$ of $\tilde{K}'$. The contour deformation relates the residues. For the $x^0<0$ choice used above, the energy integral can be written schematically as

$$I_{M(x^0)} = -i\int_{\text{F.C.}}\frac{\mathrm{d}z'}{2\pi}\frac{e^{-ix^0z'}}{\tilde{K}'(z')}. \tag{337}$$

The precise contour specifies how each real-axis pole is bypassed. The Euclidean-to-Minkowski momentum substitution is

$$p_E^0 = -ip^0, \quad p_E^i = -p^i, \quad p_{E,i} = p_i, \quad x_E^0 = ix^0, \quad x_E^i = x^i. \tag{338}$$

Consequently, $p_E^2 \to -p^2$ and $p_E\cdot x_E \to p\cdot x$. For the scalar,

$$\frac{1}{p_E^2+m^2} \to \frac{1}{-p^2+m^2}. \tag{339}$$

The measure and contour rotation supply the additional factor that gives the Minkowski propagator. With the convention used here,

$$\Delta_{F(x)} = \int\frac{\mathrm{d}^4p}{(2\pi)^4}\frac{ie^{-ip\cdot x}}{p^2-m^2+i\varepsilon}. \tag{340}$$

The infinitesimal $i\varepsilon$ encodes the same pole bypass as the Feynman contour: the positive-energy pole lies just below, and the negative-energy pole just above, the real $p^0$ axis.

Equivalently, one may keep the infinitesimal displacement of the poles in a specified Feynman contour integral. The Euclidean energy and time are related to their Minkowski counterparts by

$$p_E^0 = -ip^0, \quad x_E^0 = ix^0, \quad p_E^2 = -p^2. \tag{341}$$

The spatial covariant components remain the same under the chosen convention, while the contravariant spatial components change sign. The general prescription is to continue a Euclidean momentum integral by deforming the energy line into the Feynman contour while avoiding poles; one does not merely replace $p_E^0$ in an unevaluated integral and forget the contour.

## 4.5 Interaction vertices

For $S_E = S_0 + S_{\text{int}}$, the interacting Euclidean correlator is

$$\langle \varphi(x_1)...\varphi(x_n) \rangle_E = \frac{\int \mathcal{D}\varphi \varphi(x_1)...\varphi(x_n) e^{-S_0 - S_{\text{int}}}}{\int \mathcal{D}\varphi e^{-S_0 - S_{\text{int}}}} = \frac{\left\langle \varphi(x_1)...\varphi(x_n) e^{-S_{\text{int}}} \right\rangle_0}{\left\langle e^{-S_{\text{int}}} \right\rangle_0}. \tag{342}$$

Expand the interaction:

$$e^{-S_{\text{int}}} = \sum_{k=0}^{\infty} \frac{\left(-S_{\text{int}}\right)^k}{k!}. \tag{343}$$

For example, the unnormalised numerator of an interacting correlator is

$$\int \mathcal{D}\varphi \varphi(x_1)...\varphi(x_n) e^{-S_0} \sum_{k=0}^{\infty} \frac{\left(-S_{\text{int}}\right)^k}{k!}. \tag{344}$$

Each term is a Gaussian expectation value of fields and can be evaluated with Wick contractions. The denominator removes the vacuum diagrams disconnected from all operator insertions.

The relation to canonical perturbation theory is especially transparent in the interaction picture. With the usual identification of the interaction part,

$$\int \mathrm{d}t H_{I(t)} = -S_{\text{int}}^M = -\int \mathrm{d}t L_{\text{int}(t)}, \tag{345}$$

the Wick-rotated expansion corresponds to the Dyson expansion of time-ordered products. For derivative interactions, this identification requires care in canonical language; the path-integral expansion is defined directly from the action.

The source method packages the perturbative expansion into

$$Z[J] = \int \mathcal{D}\varphi e^{-S_0[\varphi] - S_{\text{int}[\varphi]} + \int \mathrm{d}^4 x J(x)\varphi(x)} = \exp -S_{\text{int}\left[\frac{\delta}{\delta J}\right]} Z_0[J]. \tag{346}$$

More explicitly,

$$Z[J] = \sum_{k=0}^{\infty} \frac{1}{k!} -S_{\text{int}\left[\frac{\delta}{\delta J}\right]}{}^k Z_0[J]. \tag{347}$$

For a field insertion, $\varphi(x)$ is replaced by $\frac{\delta}{\delta J(x)}$. This is a formal identity understood through its power series. As in canonical perturbation theory, insertions of $S_{\text{int}}$ generate interaction vertices.

At first order,

$$Z^1[J] = -S_{\text{int}\left[\frac{\delta}{\delta J}\right]} Z_0[J]. \tag{348}$$

When every derivative in $S_{\text{int}\left[\frac{\delta}{\delta J}\right]}$ acts on the exponential rather than on a preceding $D \cdot J$, its contribution is

$$Z^1_{\text{selected}}[J] = -Z_0[0] S_{\text{int}[D \cdot J]} e^{\frac{1}{2} J \cdot D \cdot J}. \tag{349}$$

The subscript selected'' refers to this one contribution; the other terms contain contractions among fields in the interaction itself.

Writing $Z_0[J] = Z_0[0]e^{J\cdot D\cdot\frac{J}{2}}$, one derivative acting on the exponential produces

$$\frac{\delta}{\delta J(x)}e^{J\cdot D\cdot\frac{J}{2}} = (D\cdot J)(x)e^{J\cdot D\cdot\frac{J}{2}}, \quad (D\cdot J)(x) = \int \mathrm{d}^4 y D_{xy}J(y). \tag{350}$$

Further derivatives can also act on the factors $D\cdot J$. These terms are internal contractions. For the connected $L$-point function at first order, the term with all $L$ external legs attached to the interaction is

$$\frac{\delta}{\delta J(x)}\int \mathrm{d}^4 y D_{zy}J(y) = D_{zx}. \tag{351}$$

For example, two derivatives acting at the same interaction point can contract with each other, producing $D_{xx}$. In diagrams this is a loop at the vertex. When one seeks the basic $L$-leg vertex, one retains the term where each of the $L$ source derivatives supplies a distinct external leg.

The first-order connected correlator from that term is

$$\langle\varphi(x_1)...\varphi(x_L)\rangle^1_{\text{conn}} = -\int\prod_{j=1}^{L}\mathrm{d}^4 y_j \frac{\delta^L S_{\text{int}[\varphi]}}{\delta\varphi(y_1)...\delta\varphi(y_L)}_{\varphi=0}\prod_{j=1}^{L} D_{x_j y_j}. \tag{352}$$

The functional derivative of the interaction is therefore the amputated vertex factor, with external propagators supplied separately.

If $S_{\text{int}}$ contains a monomial with exactly $L$ fields, its $L$-th functional derivative evaluated at $\varphi = 0$ supplies the contact vertex. A monomial with fewer fields cannot supply $L$ distinct external legs at this order; a monomial with additional fields may also generate internal contractions, but those belong to other diagrams.

### 4.5.1 Local $\varphi^4$ example

For the Minkowski interaction $\mathcal{L}_{\text{int}} = -\lambda\frac{\varphi^4}{4!}$, Wick rotation gives

$$S^E_{\text{int}} = -iS^M_{\text{int}}|_{t=-i\tau} = -i-\frac{\lambda}{4!}\int \mathrm{d}^4 x\varphi(x)^4_{\;\;t=-i\tau} = \frac{\lambda}{4!}\int \mathrm{d}^4 y\varphi(y)^4. \tag{353}$$

Thus

$$S^E_{\text{int}} = \frac{\lambda}{4!}\int \mathrm{d}^4 y\varphi(y)^4. \tag{354}$$

Four functional derivatives give

$$\frac{\delta^4 S^E_{\text{int}}}{\delta\varphi(y_1)\delta\varphi(y_2)\delta\varphi(y_3)\delta\varphi(y_4)}_{\varphi=0} = \lambda\int \mathrm{d}^4 y\prod_{j=1}^{4}\delta^4(y-y_j). \tag{355}$$

The connected first-order correlator is

$$\langle\varphi(x_1)\varphi(x_2)\varphi(x_3)\varphi(x_4)\rangle^1_{\text{conn}} = -\lambda\int \mathrm{d}^4 y\prod_{j=1}^{4} D_{x_j y}. \tag{356}$$

The four delta functions in the functional derivative set all field positions at the same vertex position $y$; only its one spacetime integral remains. The 4! ways to attach the four identical fields cancel the $\frac{1}{4!}$ in the action.

Thus the Euclidean amputated vertex is $-\lambda$. Continuing to Minkowski signature gives the familiar vertex factor $-i\lambda$.

### 4.5.2 Derivative interactions in momentum space

With the Fourier convention above, each derivative on a field of incoming momentum $p$ produces $-ip_\mu$. A local interaction takes the form

$$S_{\mathrm{int}[\varphi]} = \int \prod_{j=1}^{L} \frac{\mathrm{d}^4 p_j}{(2\pi)^4} (2\pi)^4 \delta^4 \sum_{j=1}^{L} p_j \mathcal{I}(p_1, ..., p_L) \prod_{j=1}^{L} \tilde{\varphi}(p_j), \tag{357}$$

where $\mathcal{I}$ contains the couplings and the momentum factors from derivatives. All momenta are taken as incoming. The delta function enforces momentum conservation. The vertex is obtained by differentiating the interaction with respect to its momentum-space fields; the external propagators are then removed.

To see the delta function explicitly, Fourier transform every external propagator. For each attachment to a vertex at $y_j$ one gets an integral of the form

$$\int \mathrm{d}^4 y_j e^{ik_j \cdot y_j} e^{-iq_j \cdot y_j} = (2\pi)^4 \delta^4(k_j - q_j). \tag{358}$$

It removes the corresponding internal momentum integral. The remaining vertex-position integral produces

$$\int \mathrm{d}^4 y e^{-i(k_1 + ... + k_L)\cdot y} = (2\pi)^4 \delta^4(k_1 + ... + k_L). \tag{359}$$

Consequently the Fourier-space connected correlator at first order has external propagator factors and an amputated interaction factor:

$$\tilde{G}^1_L(k_1, ..., k_L) = \prod_{j=1}^{L} \frac{1}{\tilde{K}(k_j)} \mathcal{V}_{E(k_1,...,k_L)} (2\pi)^4 \delta^4 \sum_{j=1}^{L} k_j, \tag{360}$$

$$\mathcal{V}_{E(k_1,...,k_L)} = -\frac{\delta^L S^E_{\mathrm{int}}}{\delta\tilde{\varphi}(k_1)...\delta\tilde{\varphi}(k_L)}_{\text{without the momentum delta}}. \tag{361}$$

The Minkowski vertex is obtained by the same continuation as the propagator. Because a local vertex has no propagator pole, its energy arguments are continued directly, together with the overall Wick-rotation factor.

When several field species are present, take the derivatives with respect to the corresponding momentum-space fields, retaining their component labels and the order of their momentum arguments.

### 4.5.3 Scalar QED example

Expanding the scalar-QED kinetic term gives interactions of the schematic form

$$\mathcal{L}_{\mathrm{int}} = ieA^\mu(\varphi^\dagger \partial_\mu \varphi - (\partial_\mu \varphi^\dagger)\varphi) + e^2 A_\mu A^\mu \varphi^\dagger \varphi. \tag{362}$$

The free charged-scalar term is $\partial_\mu \varphi^\dagger \partial^\mu \varphi - m^2|\varphi|^2$. To see the momentum dependence of the interaction, write for each field

$$\varphi(x) = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} e^{-ip\cdot x} \tilde{\varphi}(p), \quad \partial_\mu \varphi(x) = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} (-ip_\mu) e^{-ip\cdot x} \tilde{\varphi}(p). \tag{363}$$

Substituting these into the cubic interaction produces the momentum-dependent factor

$$ie(-ip_+)_\mu - (-ip_-)_\mu \tilde{\varphi}^{\dagger(p_-)}\tilde{\varphi}(p_+)\tilde{A}^{\mu(k)}, \tag{364}$$

along with $(2\pi)^4\delta^4(p_+ + p_- + k)$. Here $p_+$ labels $\varphi$ and $p_-$ labels $\varphi^\dagger$ in this incoming-momentum convention. Including the overall $i$ from the Minkowski interaction insertion, the three-point vertex factor is

$$ie(p_+ - p_-)_\mu, \tag{365}$$

and the two-photon, two-scalar contact vertex is

$$2ie^2\eta_{\mu\nu}. \tag{366}$$

The momentum factors arise from the derivatives in the interaction, while the factor of two in the contact vertex comes from differentiating twice with respect to the photon field.

# Bibliography